Experiments

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Abstract

We consider a Random Weight Neural Network (RWNN) regression model and wish to make predictions based on estimating posterior distributions of weights and biases and hence sampling.

1 Setup

Let $\{(Y_i, X_i)\}_{i=1}^n$, with $n \in \mathbb{N}$, $Y_i \in \mathbb{R}$ and $X_i \in \mathbb{R}$, be data sampled i.i.d. according to the nonparametric regression model

$$Y_i = h(X_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma_\epsilon^2), \tag{1}$$

where $h : \mathbb{R} \to \mathbb{R}$ is an unknown function.

Random Weight Neural Network. We approximate h using a Random Weight Neural Network (RWNN):

$$Y_i = \sum_{k=1}^{K} w_k \, \rho(a_k X_i + b_k) + \epsilon_i, \tag{2}$$

where $\rho = \tanh$; $a_k \sim \mathcal{N}(0, \gamma_a^2)$ and $b_k \sim \mathcal{N}(0, \gamma_b^2)$ are the weights and biases, randomly sampled independently of the data; and w_k are the readout coefficients. In matrix notation,

$$Y = \rho\big(XA^\top + B^\top\big)\, W + \epsilon, \tag{3}$$

where $Y \in \mathbb{R}^n$, $X \in \mathbb{R}^n$, $W \in \mathbb{R}^K$, $A \in \mathbb{R}^K$, $B \in \mathbb{R}^K$, and $\epsilon \in \mathbb{R}^n$. The RWNN feature matrix is defined as

$$F_{A,B} := \rho\big(XA^\top + B^\top\big) \in \mathbb{R}^{n \times K}. \tag{4}$$

Ridge Regression for W. Given fixed A and B, we estimate W by ridge-regularised least squares:

$$\hat{W}_\lambda := \arg\min_{W \in \mathbb{R}^K} \left\{ \|Y - F_{A,B} W\|_2^2 + \lambda \|W\|_2^2 \right\}, \tag{5}$$

which admits the closed-form solution

$$\hat{W}_\lambda = \left(F_{A,B}^\top F_{A,B} + \lambda I_K\right)^{-1} F_{A,B}^\top Y. \tag{6}$$

This estimator coincides with the posterior mean of the Bayesian linear regression under priors $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2 I_n)$ and $W \sim \mathcal{N}(0, (\lambda/\tau) I_K)$, where $\tau = 1/\sigma_\epsilon^2$. The resulting posterior is

$$W \mid Y, X, A, B, \sigma_\epsilon^2 \sim \mathcal{N}\left(\mu_n,\, (\tau \Lambda_n)^{-1}\right), \tag{7}$$

where $\mu_n := \Lambda_n^{-1} F_{A,B}^\top Y$ and $\Lambda_n := F_{A,B}^\top F_{A,B} + \lambda I_K$.

2 Computing Posterior Distributions of A and B

2.1 Intuition

We may consider approximating densities of A and B rather than giving a point estimate of them since estimating A,B and W can be done very efficiently in more complex NNs. And we can estimate W given A and B easily. One direction here is to calculate posterior densities analytically. The key idea is to do something efficient via sampling and avoid estimation of the parameters. We could try to estimate densities by Tractable Approximate Gaussian Inference(TAGI). This method is plausible as we consider the prior distributions of A and B as Gaussian.

2.2 Procedure

The following method is a simplified version of that in `https://www.jmlr.org/papers/volume22/20-1009/20-1009.pdf`. as our model has only one layer.

2.2.1 New Setting

For observation i and hidden unit k, define

$$Z_{ik} = a_k X_i + b_k, \tag{8}$$

and

$$\Phi_{ik} = \tanh(Z_{ik}). \tag{9}$$

The model output before adding observation noise is therefore

$$S_i = \sum_{k=1}^{K} w_k \Phi_{ik}, \tag{10}$$

such that

$$Y_i = S_i + \epsilon_i. \tag{11}$$

Also, for computational simplicity, we adopt the diagonal covariance approximation used in the original TAGI framework. In particular, we assume

$$\mathrm{Cov}(a_k, b_l) = 0 \tag{12}$$

for all k, l, and

$$\mathrm{Cov}(a_k, a_l) = \mathrm{Cov}(b_k, b_l) = 0, \qquad k \neq l. \tag{13}$$

2.2.2 Initial Ridge Estimate of W

The TAGI update of A and B requires a value of the readout coefficient vector W. We first sample initial hidden parameters from their priors:

$$A^{(0)} \sim N(0, \gamma_a^2 I_K), \qquad B^{(0)} \sim N(0, \gamma_b^2 I_K). \tag{14}$$

Using these initial values, we construct the feature matrix

$$F_{ik}^{(0)} = \tanh\left(a_k^{(0)} X_i + b_k^{(0)}\right). \tag{15}$$

An initial estimate of the readout coefficients is then obtained by ridge regression:

$$W^{(0)} = \left(F^{(0)\top} F^{(0)} + \lambda I_K\right)^{-1} F^{(0)\top} Y. \tag{16}$$

The vector $W^{(0)}$ is held fixed during the subsequent updates.

2.2.3 Forward Uncertainty Propagation

Suppose that, before observing Y_i,

$$a_k \sim N(\mu_{a_k}, \sigma_{a_k}^2); \qquad b_k \sim N(\mu_{b_k}, \sigma_{b_k}^2). \tag{17}$$

Because X_i is treated as fixed and $\mathrm{Cov}(a_k, b_k) = 0$, the linear hidden variable in Equation (8) has mean

$$\mu_{Z_{ik}} = X_i \mu_{a_k} + \mu_{b_k} \tag{18}$$

and variance

$$\sigma_{Z_{ik}}^2 = X_i^2 \sigma_{a_k}^2 + \sigma_{b_k}^2. \tag{19}$$

The non-linear transformation $\Phi_{ik} = \tanh(Z_{ik})$ does not preserve Gaussianity. According to the paper above, we apply a local linearisation of the activation function around $\mu_{Z_{ik}}$:

$$\tanh(Z_{ik}) \approx \tanh(\mu_{Z_{ik}}) + J_{ik}\left(Z_{ik} - \mu_{Z_{ik}}\right), \tag{20}$$

where

$$J_{ik} = \left.\frac{\partial \tanh(z)}{\partial z}\right|_{z=\mu_{Z_{ik}}} = 1 - \tanh^2(\mu_{Z_{ik}}). \tag{21}$$

The approximate activation mean and variance are consequently

$$\mu_{\Phi_{ik}} \approx \tanh(\mu_{Z_{ik}}) \tag{22}$$

and

$$\sigma_{\Phi_{ik}}^2 \approx J_{ik}^2 \sigma_{Z_{ik}}^2. \tag{23}$$

2.2.4 Predictive Mean and Variance

Conditional on $W^{(0)}$, the predictive mean for observation i is

$$\mu_{Y_i} = \sum_{k=1}^{K} w_k^{(0)} \mu_{\Phi_{ik}}. \tag{24}$$

Under the diagonal covariance approximation, the predictive variance is

$$S_i := \mathrm{Var}(Y_i) \approx \sum_{k=1}^{K} \left(w_k^{(0)}\right)^2 \sigma_{\Phi_{ik}}^2 + \sigma_\epsilon^2. \tag{25}$$

2.2.5 Evaluation of Covariances

The covariances between the hidden parameters and Z_{ik} are

$$\mathrm{Cov}(a_k, Z_{ik}) = X_i \sigma_{a_k}^2 \tag{26}$$

and

$$\mathrm{Cov}(b_k, Z_{ik}) = \sigma_{b_k}^2. \tag{27}$$

Using the local linearisation in Equation (20), we obtain

$$\mathrm{Cov}(a_k, \Phi_{ik}) \approx J_{ik} X_i \sigma_{a_k}^2 \tag{28}$$

and

$$\mathrm{Cov}(b_k, \Phi_{ik}) \approx J_{ik} \sigma_{b_k}^2. \tag{29}$$

Therefore, conditional on $W^{(0)}$,

$$C_{a_k Y_i} := \mathrm{Cov}(a_k, Y_i) \approx w_k^{(0)} J_{ik} X_i \sigma_{a_k}^2 \tag{30}$$

and

$$C_{b_k Y_i} := \mathrm{Cov}(b_k, Y_i) \approx w_k^{(0)} J_{ik} \sigma_{b_k}^2. \tag{31}$$

2.2.6 Posterior Update

The TAGI update is based on equation (13) of the paper above. For a general parameter vector θ, these equations are

$$\mu_{\theta|y_i} = \mu_\theta + \Sigma_{\theta Y_i} \Sigma_{Y_i}^{-1} \left(y_i - \mu_{Y_i}\right) \tag{32}$$

and

$$\Sigma_{\theta|y_i} = \Sigma_\theta - \Sigma_{\theta Y_i} \Sigma_{Y_i}^{-1} \Sigma_{Y_i \theta}. \tag{33}$$

Because Y_i is scalar, $\Sigma_{Y_i} = S_i$. Hence, the posterior mean updates for a_k and b_k are

$$\mu_{a_k}^+ = \mu_{a_k} + \frac{C_{a_k Y_i}}{S_i} \left(y_i - \mu_{Y_i}\right) \tag{34}$$

and

$$\mu_{b_k}^{+} = \mu_{b_k} + \frac{C_{b_k Y_i}}{S_i} \left(y_i - \mu_{Y_i}\right). \tag{35}$$

The corresponding marginal variance updates are

$$\left(\sigma_{a_k}^2\right)^{+} = \sigma_{a_k}^2 - \frac{C_{a_k Y_i}^2}{S_i} \tag{36}$$

and

$$\left(\sigma_{b_k}^2\right)^{+} = \sigma_{b_k}^2 - \frac{C_{b_k Y_i}^2}{S_i}. \tag{37}$$

Equations (34)–(37) are applied sequentially for $i = 1, \ldots, n$. After each observation, the updated Gaussian approximation becomes the prior approximation for the next observation.

2.2.7 Posterior Sampling

For $s = 1, \ldots, S$, we independently sample

$$a_k^{(s)} \sim N(\mu_{a_k}, \sigma_{a_k}^2), \qquad b_k^{(s)} \sim N(\mu_{b_k}, \sigma_{b_k}^2). \tag{38}$$

Each posterior sample defines a feature matrix

$$F_{ik}^{(s)} = \tanh\left(a_k^{(s)} X_i + b_k^{(s)}\right). \tag{39}$$

Conditional on $A^{(s)}$ and $B^{(s)}$, the corresponding readout coefficient vector is computed by ridge regression:

$$W^{(s)} = \left(F^{(s)\top} F^{(s)} + \lambda I_K\right)^{-1} F^{(s)\top} Y. \tag{40}$$

3 Experiments

3.1 Setup

In this section, we aim to test the prediction results for different h(x). For each of them, we simulate 1000 pairs of observations and divide them into two sets: 800 for the training set and 200 for the prediction set.

Apart from that, we are interested in how different values of λ will affect the posterior distributions and our prediction results since the value of W depends on λ.

3.2 $h(x) = \sin(x)$

We fix $K = 20$, $\gamma_a^2 = \gamma_b^2 = 0.6$, $\sigma_\epsilon^2 = 0.3$.

In order to select λ, we perform a grid search over 20 candidate values spaced logarithmically between 10^{-3} and 10^{1}. For each candidate λ, we repeat a random 80/20 train-validation split three times. On each split, we run the full TAGI procedure to evaluate the posterior predictive mean against the targets Y (obtain RMSE). We report the mean and standard deviation of RMSE for each λ, and select the value that minimizes the mean RMSE.

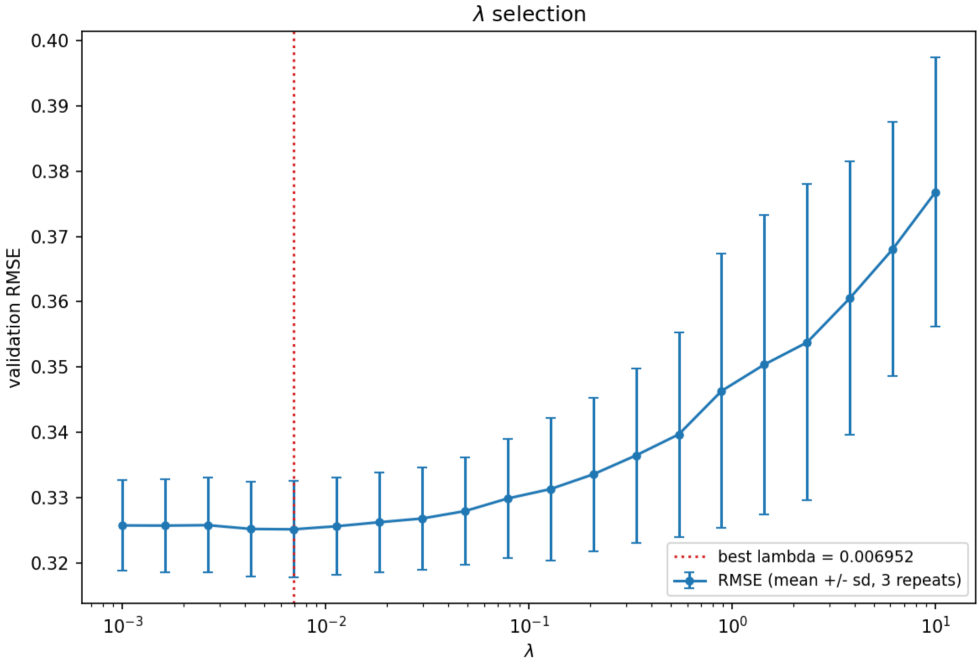


Figure 1: RMSE VS λ

Note: We use a logarithmically-spaced grid for λ since its range spans several orders of magnitude. A linear grid might neglect the small-λ region where RMSE might change significantly.

As shown in the figure, as λ increases, the mean RMSE increases. The predictions are based on sampling from posterior distributions of A and B, hence we will consider the effect of increasing λ on the mean and variance of A and B.

Clearly, the posterior variance increases as λ increases initially and becomes relatively stable for each k. This can be explained roughly as follows:

$$\left(\sigma_{a_k}^2\right)^+ = \sigma_{a_k}^2 - \frac{C_{a_k Y_i}^2}{S_i} \tag{41}$$

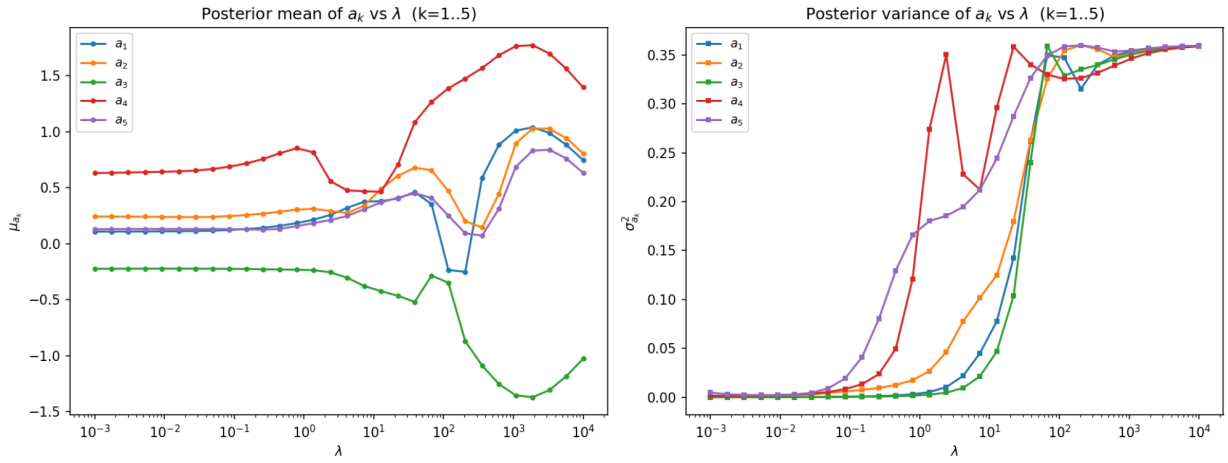
and

$$C_{a_k Y_i} := \mathrm{Cov}(a_k, Y_i) \approx w_k^{(0)} J_{ik} X_i \sigma_{a_k}^2 \tag{42}$$

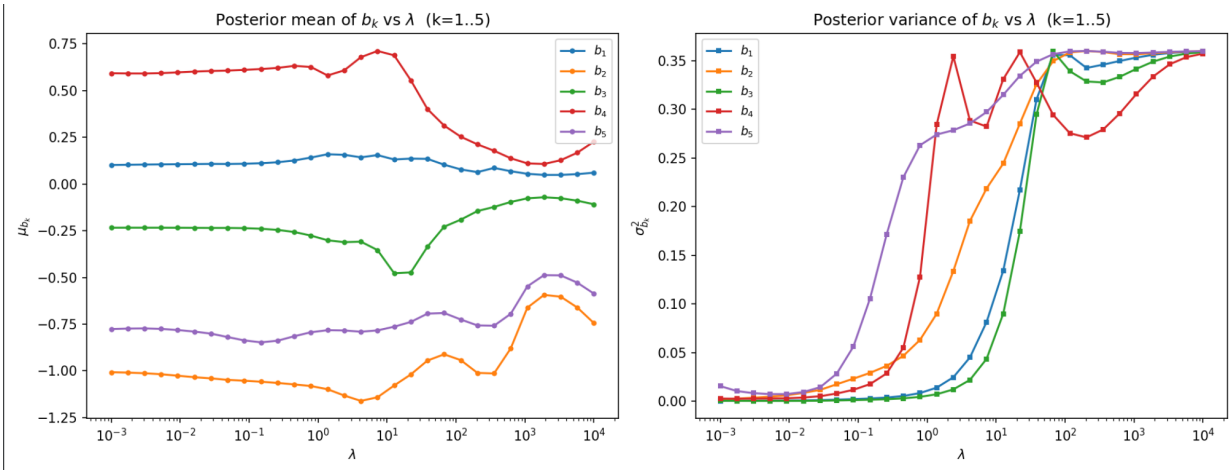
$C_{a_k Y_i}$ is directly proportional to $w_k^{(0)}$. Increasing λ pulls $W^{(0)}$ towards 0 because $W^{(0)} = \left(F^{(0)\top} F^{(0)} + \lambda I_K\right)^{-1} F^{(0)\top} Y \to 0$ as $\lambda \to \infty$.

Smaller $w_k^{(0)}$ leads to smaller $C_{a_k Y_i}$, making the variance reduction for the update step smaller. Thus, the final $\sigma_{a_k}^2$ ends up closer to the prior. This means that the data may not work well for updating posterior variance.

By contrast, the posterior means remain relatively stable for λ between 10^{-3} and 10^2.



(a) Distributions of a_k across λ



(b) Distributions of b_k across λ

The similar trend is shown for b_k. This can be explained analogously by the statement above (consider the update procedure).

Finally we can look at the posterior distributions and prediction results when $\lambda = 0.006952$.

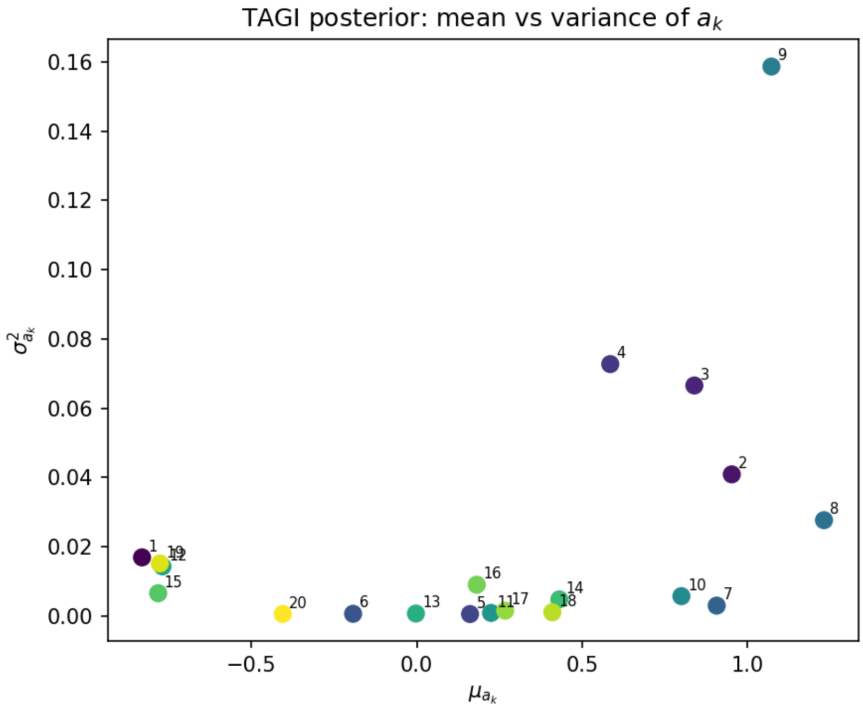


Figure 3: Posterior a_k

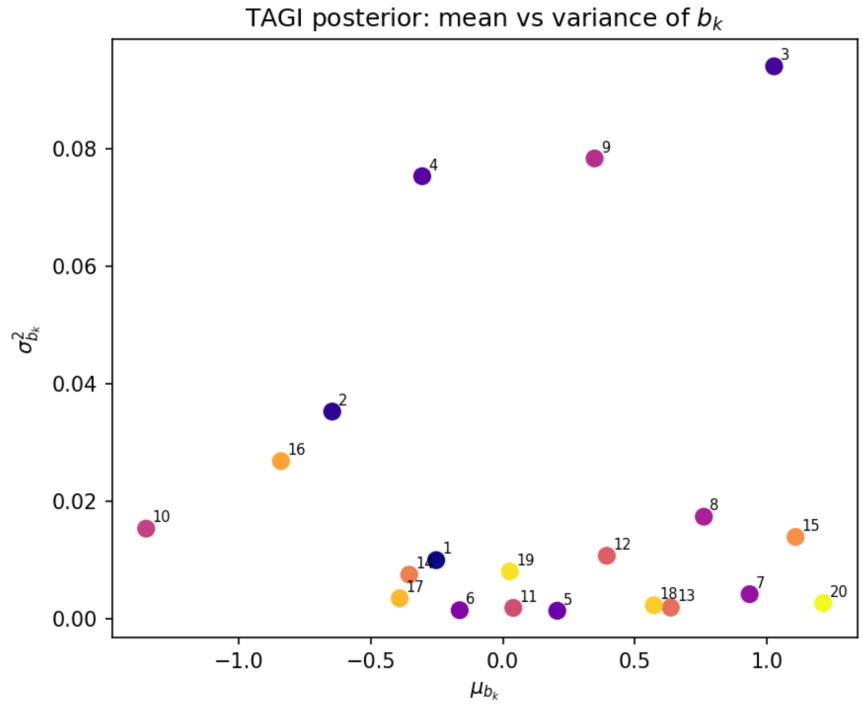


Figure 4: Posterior b_k

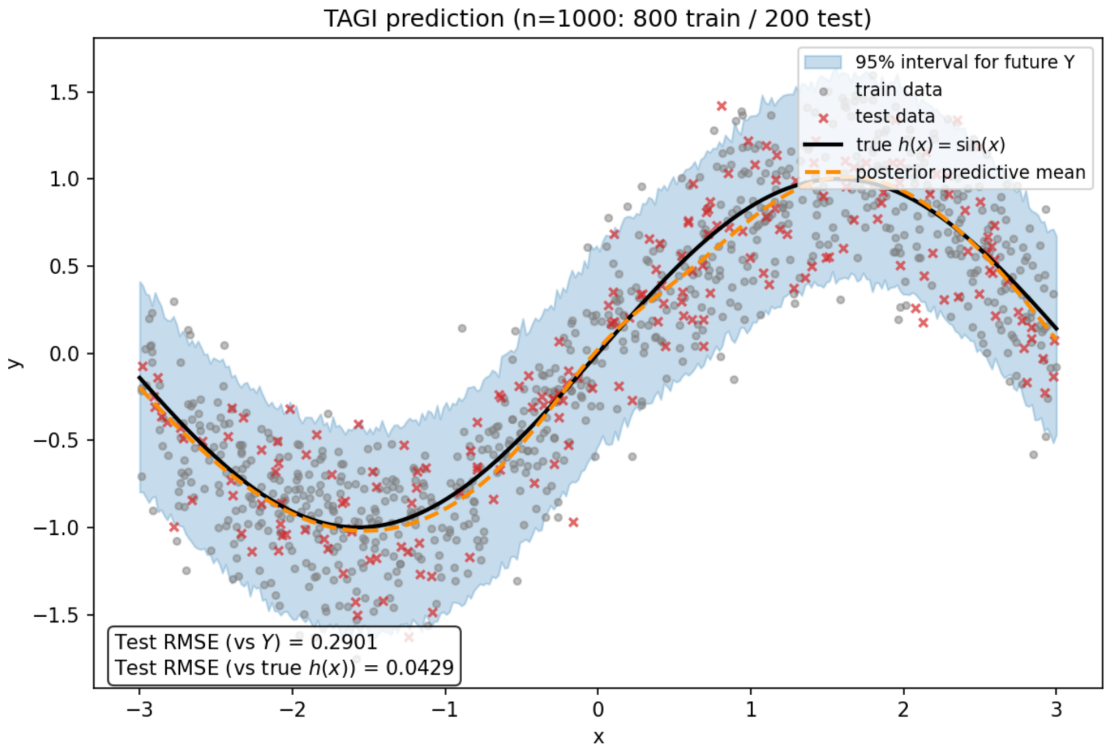


Figure 5: Prediction Results

Generally our method captures the main features of $h(x)$. 95% interval for future Y is the range within which a newly observed data is expected to fall with 95% probability.

3.3 $h(x) = x + 5\varphi(10x)$

Again we fix $K = 20$, $\gamma_a^2 = \gamma_b^2 = 0.6$, $\sigma_\epsilon^2 = 0.3$. We implement the same method to choose an appropriate λ.

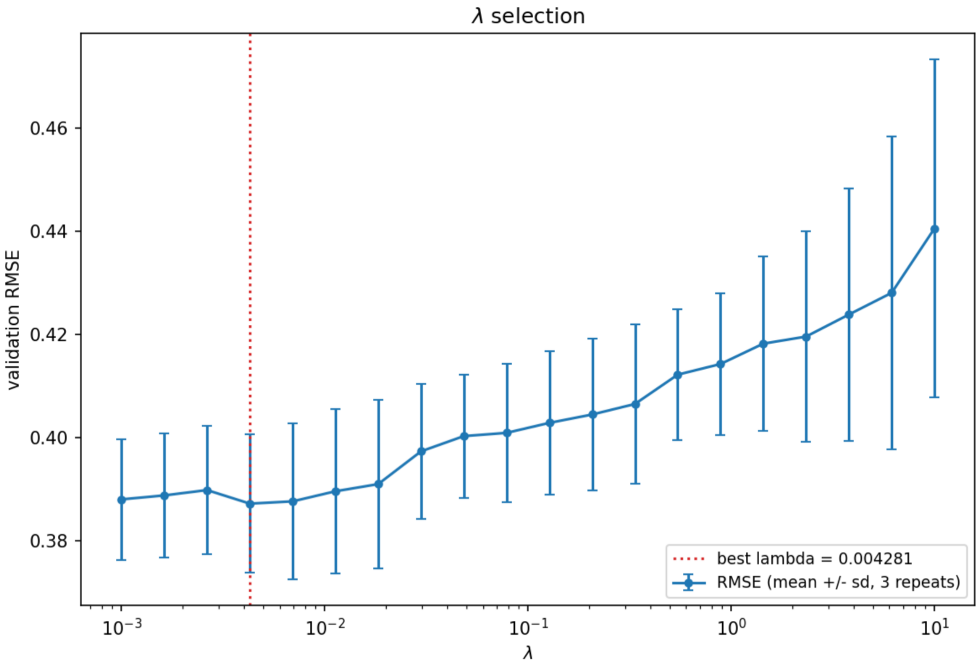


Figure 6: RMSE VS λ

By this graph, we set λ to be 0.004281. Next we can check the fitting and prediction results.

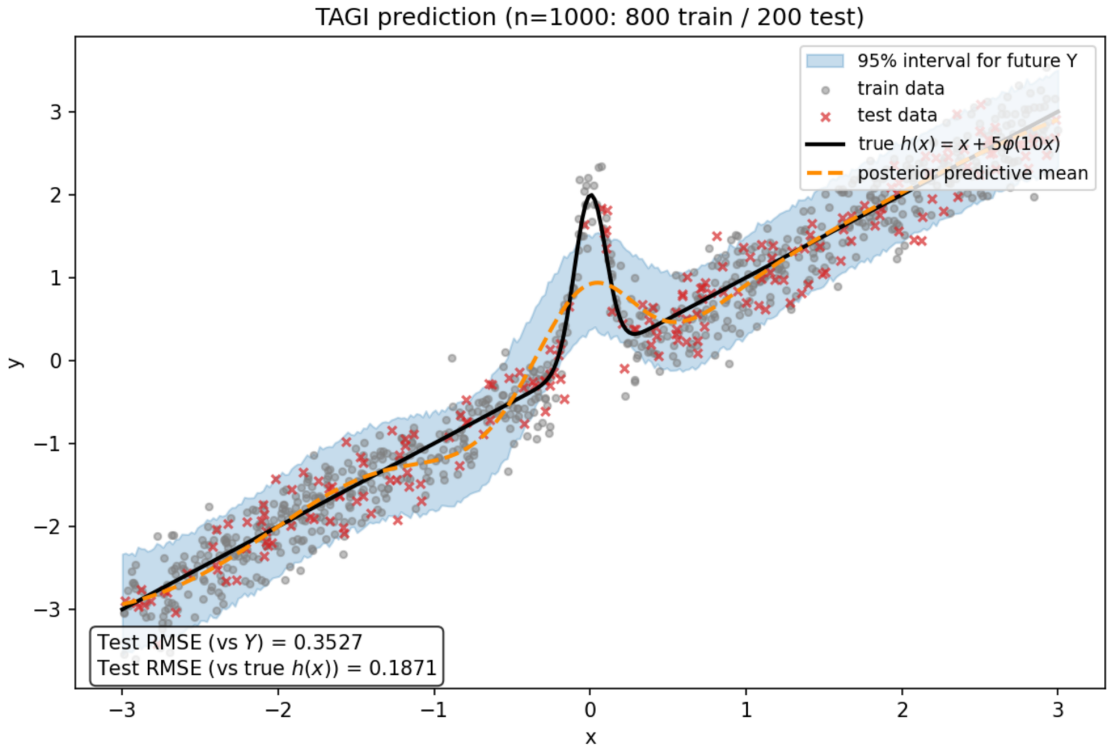


Figure 7: Initial Prediction

Around $x = 0$, $h(x)$ is sharp and our prediction curve does not capture this(its peak is much lower than the true peak).

One direction here is to adjust the hyperparameters. We could try to increase γ_a^2 and γ_b^2. Let $\gamma_a^2 = \gamma_b^2 = 3.0$. Keep other hyperparameters fixed.

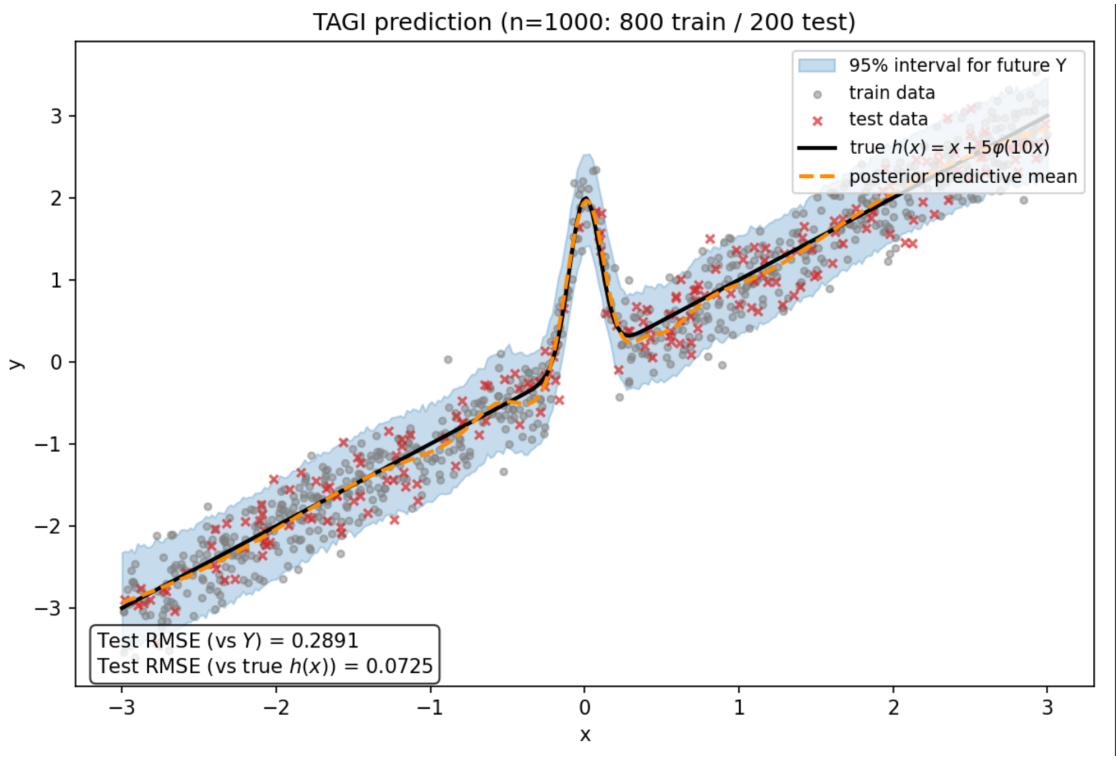


Figure 8: Improved Results

The RMSEs become much lower and the prediction curve approximately overlaps with $h(x)$. Here we are interested in why the increase in γ_a^2 and γ_b^2 could lead to such improvements.

3.4 $h(x) = \sin(3\pi x/2)/[1 + 18x^2(\text{sgn}(x) + 1)]$

Similar trend is observed in this section. Fix fix $K = 20$, $\gamma_a^2 = \gamma_b^2 = 0.6$, $\sigma_\epsilon^2 = 0.3$. Set λ to be 0.02976 by experiment.

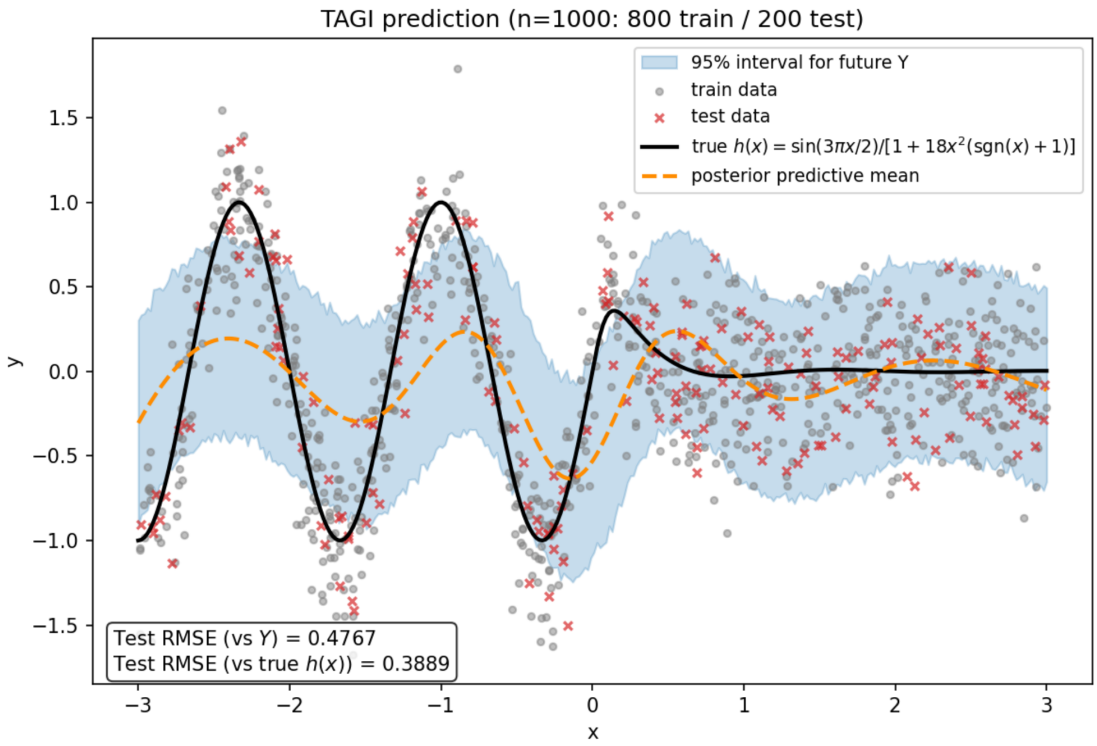


Figure 9: Initial Fitting

And then set $\gamma_a^2 = \gamma_b^2 = 3.0$.

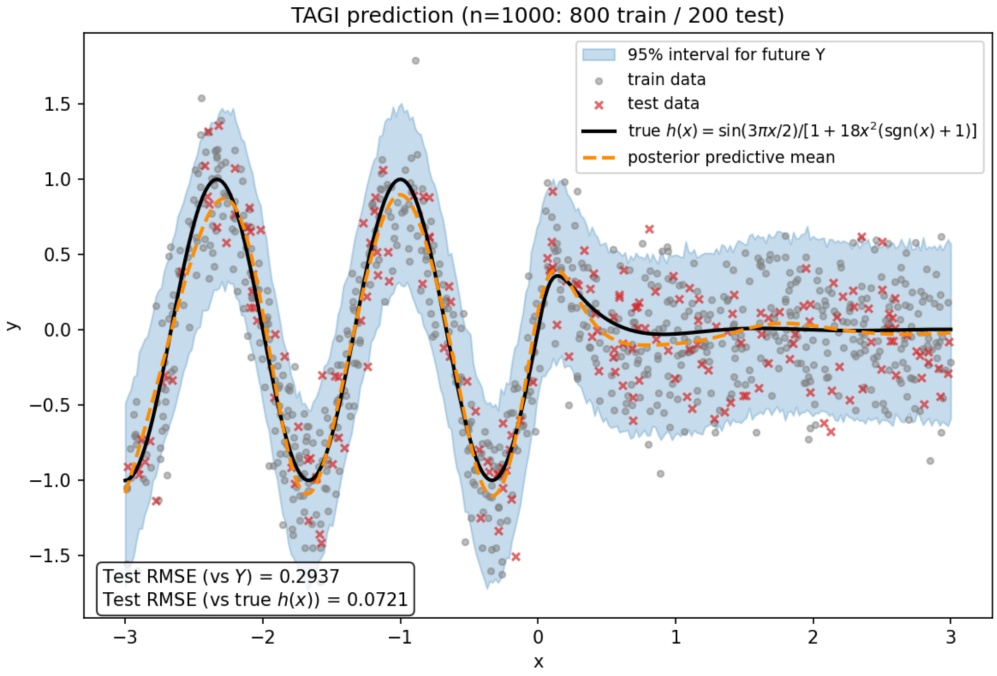


Figure 10: Improved Fitting

There is a huge improvement!

Here we may consider the capability of our model to achieve a large slope.

By the chain rule,

$$\frac{d}{dx}\tanh(a_k x + b_k) = a_k\left[1 - \tanh^2(a_k x + b_k)\right].$$

Since $\tanh^2(z) \in [0,1)$ for all $z \in \mathbb{R}$, the bracketed factor satisfies $1 - \tanh^2(z) \in (0,1]$, with equality 1 attained at $z = 0$. Hence the derivative is maximized in absolute value at the point where $a_k x + b_k = 0$, giving $\max_x |\tanh'(a_k x + b_k) \cdot a_k| = |a_k| \cdot 1 = |a_k|$. Thus For any k, $\max_x \left|\frac{d}{dx}\tanh(a_k x + b_k)\right| = |a_k|$.

Then we can show that for the fitted network $\hat{h}(x) = \sum_k w_k \tanh(a_k x + b_k)$,

$$|\hat{h}'(x)| \;\le\; \sum_{k=1}^{K} |w_k|\,|a_k| \qquad \text{for all } x.$$

This can be done easily by triangle inequality.

Intuitively, we require some $|a_k|$ to be large so that the model can achieve lagre slope. By setting γ_a^2 to be large, the simulated a_k are more likely to be large.

4 Posterior Distribution of $\widehat{h}(x)$

We want to evaluate the posterior distribution of our estimator for $h(x)$, given that we have used TAGI to obtain an approximation of the posterior for a_k and b_k. Note that this is essentially related to the derivations in Section 2.2.7 above, too.

Sampling of coefficients. For $s = 1, \ldots, S$, we independently sample

$$a_k^{(s)} \sim N(\mu_{a_k}, \sigma_{a_k}^2), \qquad b_k^{(s)} \sim N(\mu_{b_k}, \sigma_{b_k}^2), \tag{43}$$

so that we obtain $A^{(s)}$ and $B^{(s)}$. Each posterior sample defines a feature matrix

$$F_{ik}^{(s)} = \tanh\left(a_k^{(s)} X_i + b_k^{(s)}\right), \tag{44}$$

so that, conditional on $A^{(s)}$ and $B^{(s)}$, we obtain

$$W^{(s)} = \left(F^{(s)\top} F^{(s)} + \lambda I_K\right)^{-1} F^{(s)\top} Y. \tag{45}$$

Therefore, for each s, we can construct

$$\widehat{h}^{(s)}(x) := \sum_{k=1}^{K} w_k^{(s)} \tanh\left(a_k^{(s)} x + b_k^{(s)}\right). \tag{46}$$

Given some (sufficiently fine) grid of x's, we can sample $S \gg 1$ times and construct functional estimates $\{\widehat{h}^{(s)}\}_{s=1}^{S}$, and thus have a good numerical estimate of the posterior distribution $\mathbb{P}_{\widehat{h}}$ for these. Then, we can plot and compare

- Posterior mean and median of $\mathbb{P}_{\widehat{h}}$.
- 90% frequency intervals for $\mathbb{P}_{\widehat{h}}$.
- True function $h(x)$.

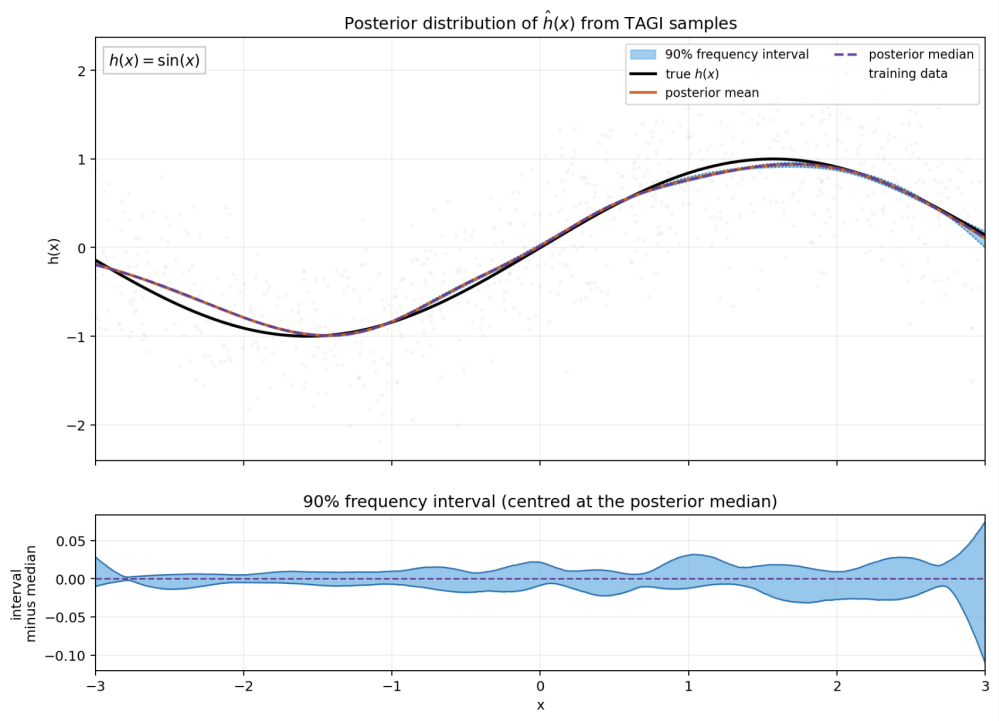


Figure 11

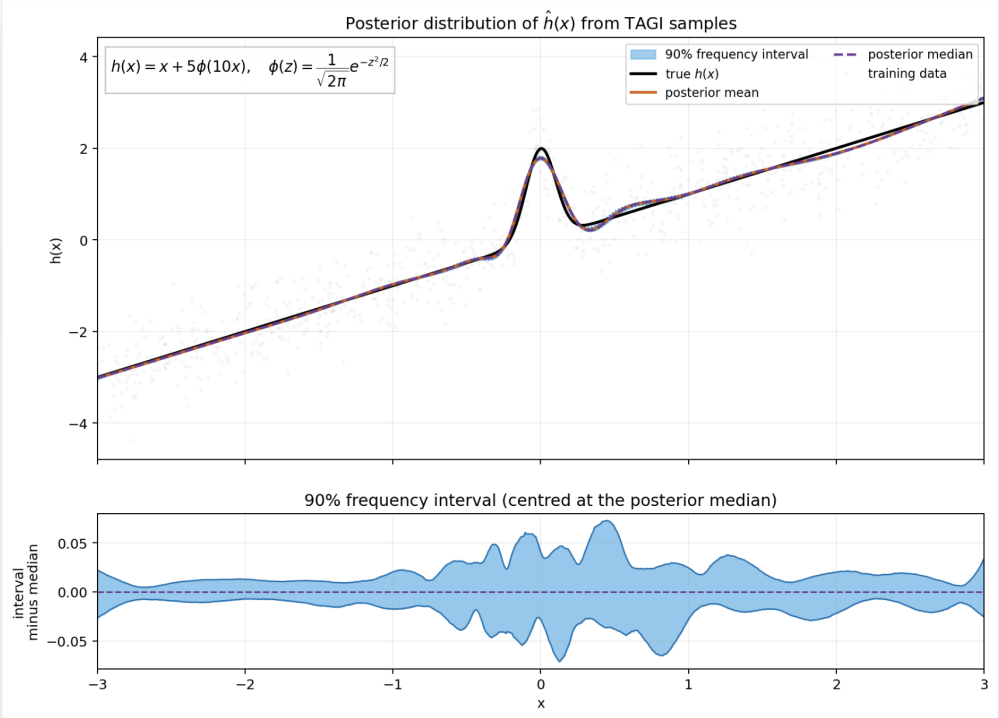


Figure 12

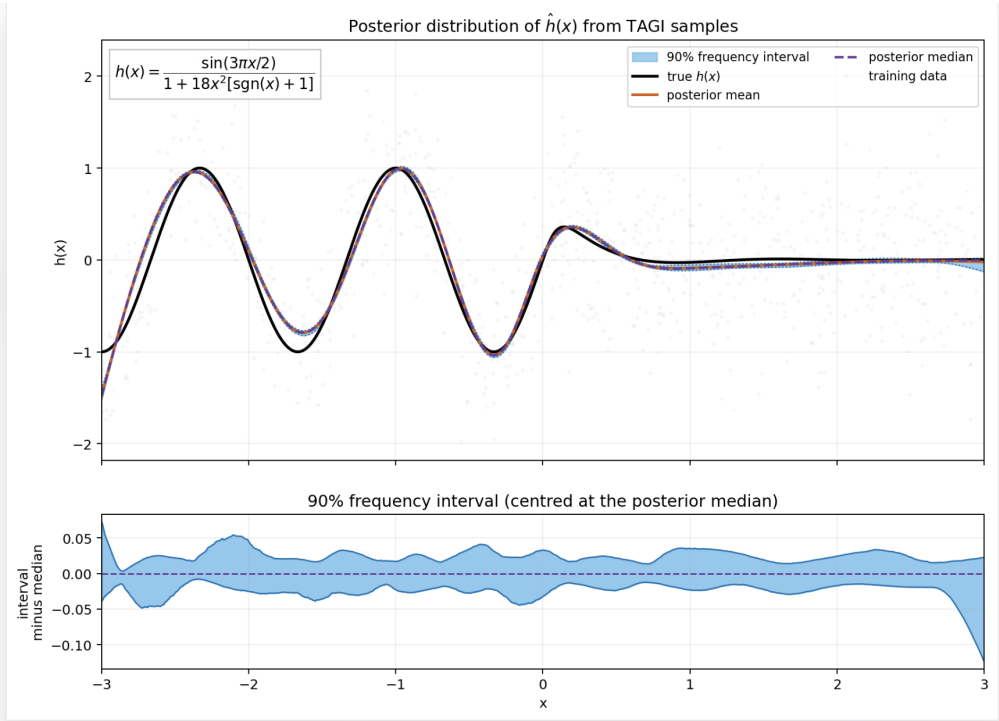


Figure 13

For all three cases, the hyperparameters remain unchanged. The posterior mean approximately overlaps with posterior median. And the frequency interval is narrow.

5 Joint Posterior Inference Using Stan

5.1 Role of Stan

To assess the accuracy of the Gaussian approximation produced by the simplified TAGI procedure, we use Stan to obtain a numerical benchmark for the joint posterior distributions of A, B and W. Stan uses Hamiltonian Monte Carlo (HMC) and No-U-Turn Sampler (NUTS)to generate samples from a target posterior distribution.

Stan does not return an analytical posterior. Instead, it provides a numerical approximation to the joint posterior without imposing the local linearisation and diagonal covariance assumptions of simplified TAGI.

Thus we can compare Stan with the simplified TAGI, but the comparison should focus on something like posterior mean, posterior median and interval width rather than marginal distribution. This is because a permutation of A, B and W leaves the output unchanged by symmetry. On the other hand, because $\tanh(\cdot)$ is an odd function,

$$w_k \tanh(a_k x + b_k) = (-w_k) \tanh(-a_k x - b_k). \tag{47}$$

Thus, the transformation

$$(a_k, b_k, w_k) \longmapsto (-a_k, -b_k, -w_k) \tag{48}$$

leaves the output unchanged. This implies that the parameters are not identifiable.

5.2 Simplified Description of Procedure

Stan generates S approximate draws from the joint posterior:

$$\left\{A^{(s)}, B^{(s)}, W^{(s)}\right\}_{s=1}^{S} \sim p(A, B, W \mid X, Y). \tag{49}$$

The samples therefore retain the posterior dependency between A, B and W unlike the diagonal covariance assumption used by the simplified TAGI.

Although Stan samples from the joint posterior, a marginal posterior distribution can be approximated by selecting the relevant coordinate from every joint draw. For example, the samples

$$\left\{a_k^{(s)}\right\}_{s=1}^{S} \tag{50}$$

provide a Monte Carlo approximation to

$$p(a_k \mid X, Y). \tag{51}$$

The marginal posterior distributions of b_k and w_k are obtained analogously.

In addition, we can obtain posterior distribution of $\widehat{h}(x)$ similarly as before.

5.3 Test with $h(x) = sin(x)$

Similarly, we set $K = 20$, $\gamma_a^2 = \gamma_b^2 = 0.6$, $\sigma_\epsilon^2 = 0.3$. In particular, we generate 2000 draws.

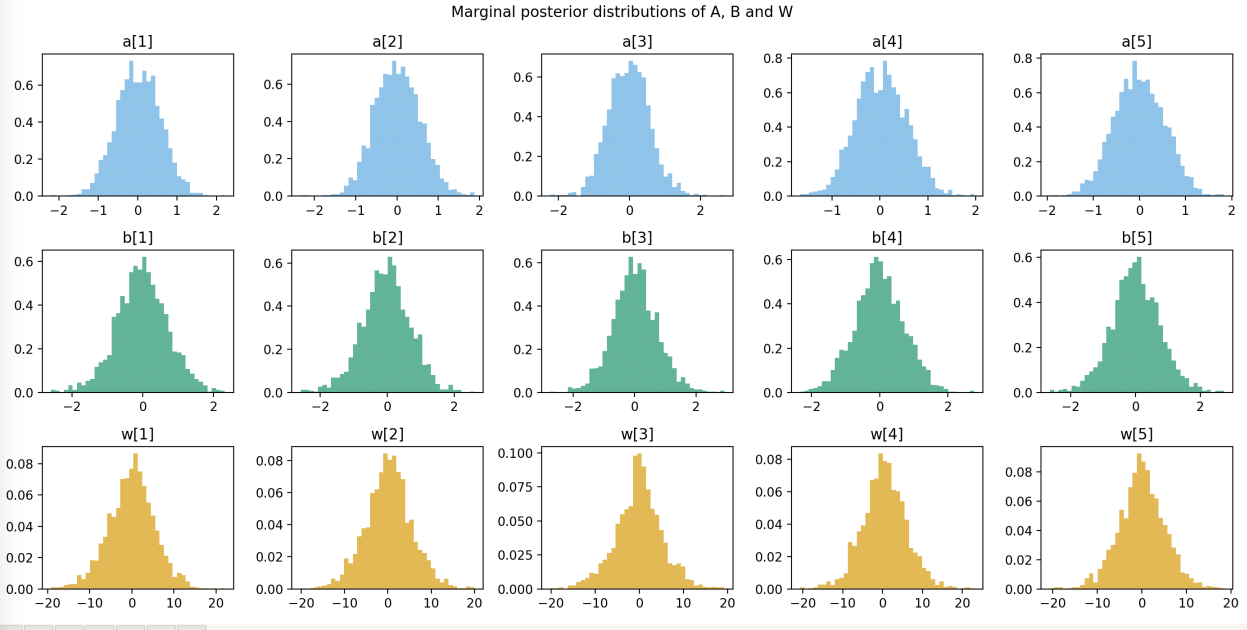


Figure 14: Marginal Distributions

The posterior marginal distributions are nearly Gaussian with mean 0, indicating that the Gaussian approximation used by TAGI is resonable. Then we can look at the posterior distribution of $\widehat{h}(x)$.

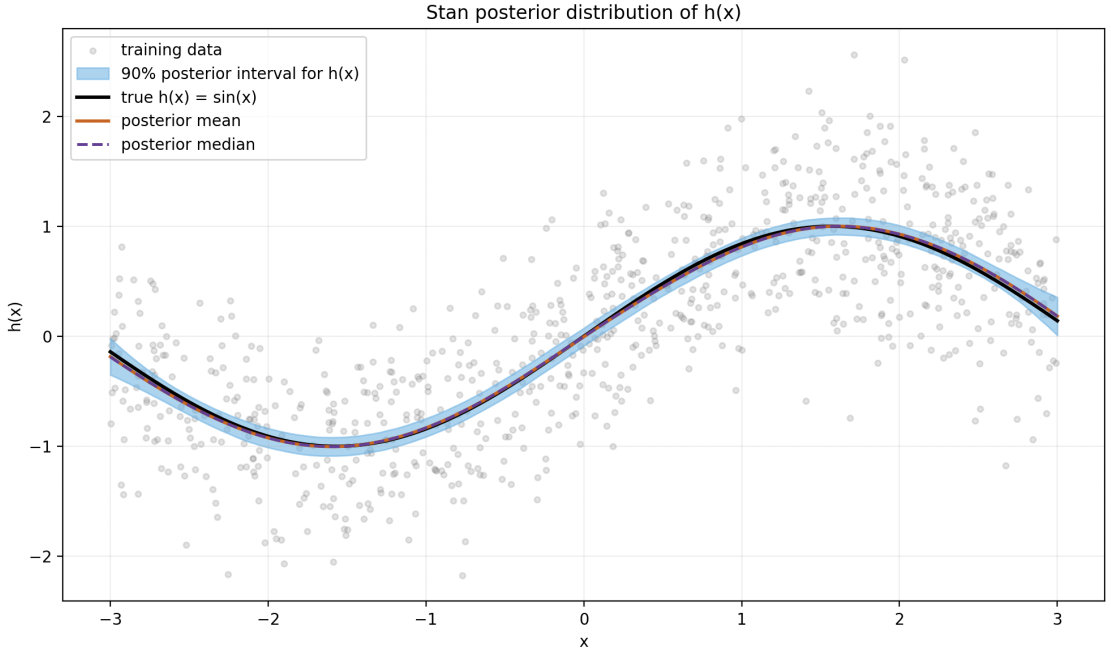


Figure 15

The posterior mean and median approximately overlap with true $h(x)$. The interval is wider than that of TAGI. So TAGI might underestimate the variability of $\widehat{h}(x)$.

5.4 Test with $h(x) = \sin(3\pi x/2)/[1 + 18x^2(\text{sgn}(x) + 1)]$

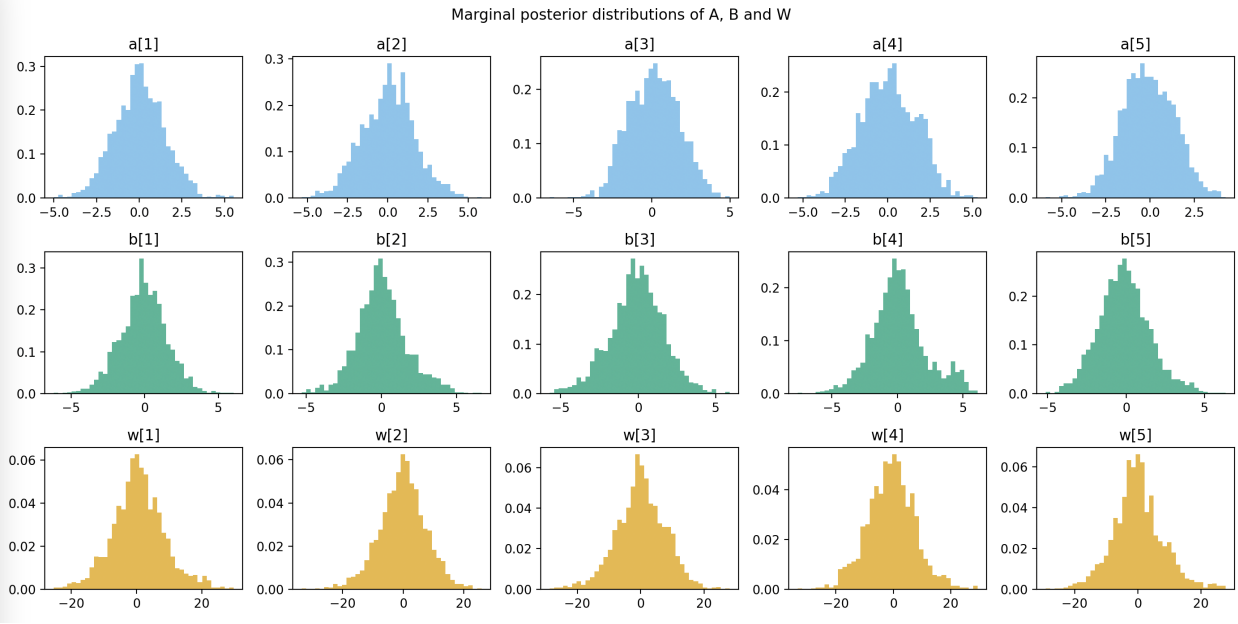


Figure 16: Marginal Distributions

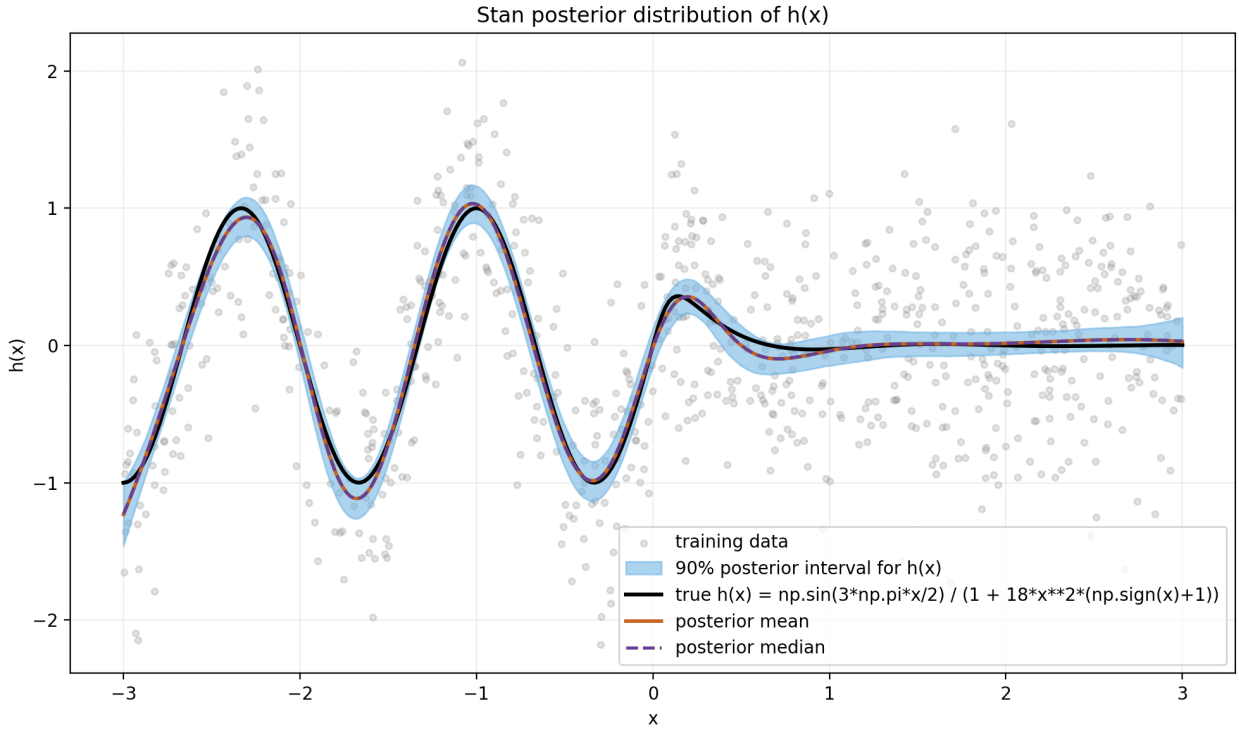


Figure 17

6 Learning Distributions of γ_a and γ_b

6.1 Motivation

In the previous analysis, γ_a and γ_b were fixed before observing the data. The resulting predictions may depend strongly on the particular values assigned to them. We therefore extend the model by treating γ_a and γ_b as unkown random variables and we wish to learn their posterior distributions from the observations.

The parameters γ_a and γ_b are defined as the standard deviations of the Gaussian distributions assigned to the hidden-layer parameters:

$$a_k \mid \gamma_a \sim \mathcal{N}(0, \gamma_a^2), \qquad k = 1, \ldots, K, \tag{52}$$

$$b_k \mid \gamma_b \sim \mathcal{N}(0, \gamma_b^2), \qquad k = 1, \ldots, K. \tag{53}$$

Since these parameters must be positive, we assign Half-Cauchy hyperpriors,

$$\gamma_a \sim \text{HalfCauchy}(0, s_a), \tag{54}$$

$$\gamma_b \sim \text{HalfCauchy}(0, s_b), \tag{55}$$

The Half-Cauchy distribution places substantial mass near zero while retaining heavy tails.

And w_k are assigned independent Gaussian priors,

$$w_k \sim \mathcal{N}(0, \tau_w^2), \qquad \tau_w = 1. \tag{56}$$

Under this hierarchical specification, the joint posterior distribution is

$$\begin{aligned} &p(A, B, W, \gamma_a, \gamma_b \mid X, Y, \sigma^2) \\ &\quad \propto p(Y \mid X, A, B, W, \sigma^2) p(A \mid \gamma_a) p(B \mid \gamma_b) p(W) p(\gamma_a) p(\gamma_b). \end{aligned} \tag{57}$$

6.2 Joint posterior sampling with Stan

We use the No-U-Turn Sampler implemented in Stan to obtain a high-accuracy reference approximation to the joint posterior in Equation (57).

A non-centred parameterisation is used to improve the geometry of the hierarchical model. Specifically, we introduce

$$z_{a,k}, z_{b,k} \overset{\text{iid}}{\sim} \mathcal{N}(0, 1) \tag{58}$$

and define

$$a_k = \gamma_a z_{a,k}, \qquad b_k = \gamma_b z_{b,k}. \tag{59}$$

Equation (59) is equivalent to the conditional priors in Equation (53), but it can reduce posterior dependence between each scale parameter and its associated hidden-layer parameters.

After discarding the warm-up iterations, the retained draws are denoted by

$$\left\{ A^{(s)}, B^{(s)}, W^{(s)}, \gamma_a^{(s)}, \gamma_b^{(s)} \right\}_{s=1}^{S}. \tag{60}$$

The samples $\{\gamma_a^{(s)}\}_{s=1}^S$ and $\{\gamma_b^{(s)}\}_{s=1}^S$ provide Monte Carlo approximations to the marginal posterior distributions $p(\gamma_a \mid X, Y)$ and $p(\gamma_b \mid X, Y)$. These distributions can be compared directly with their Half-Cauchy priors to assess how strongly the observed data update the assumed distributions.

6.3 Test with $h(x) = sin(x)$

Here we set $K = 5$ to reduce running time, $\sigma_\epsilon^2 = 0.3$ and 2000 draws again.

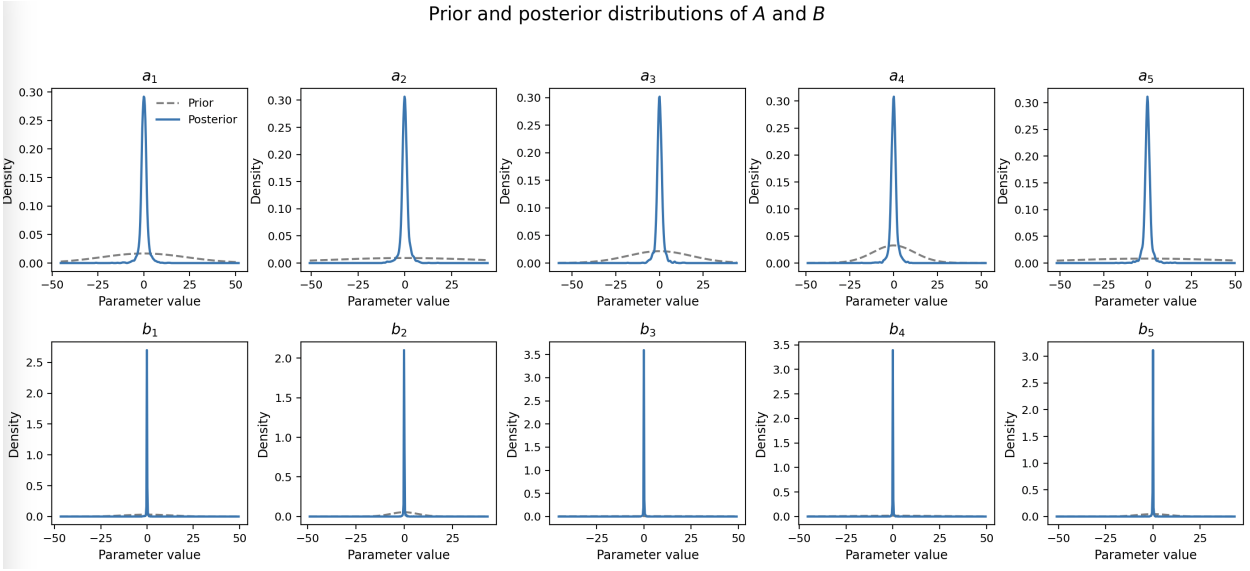


Figure 18: Prior vs Posterior

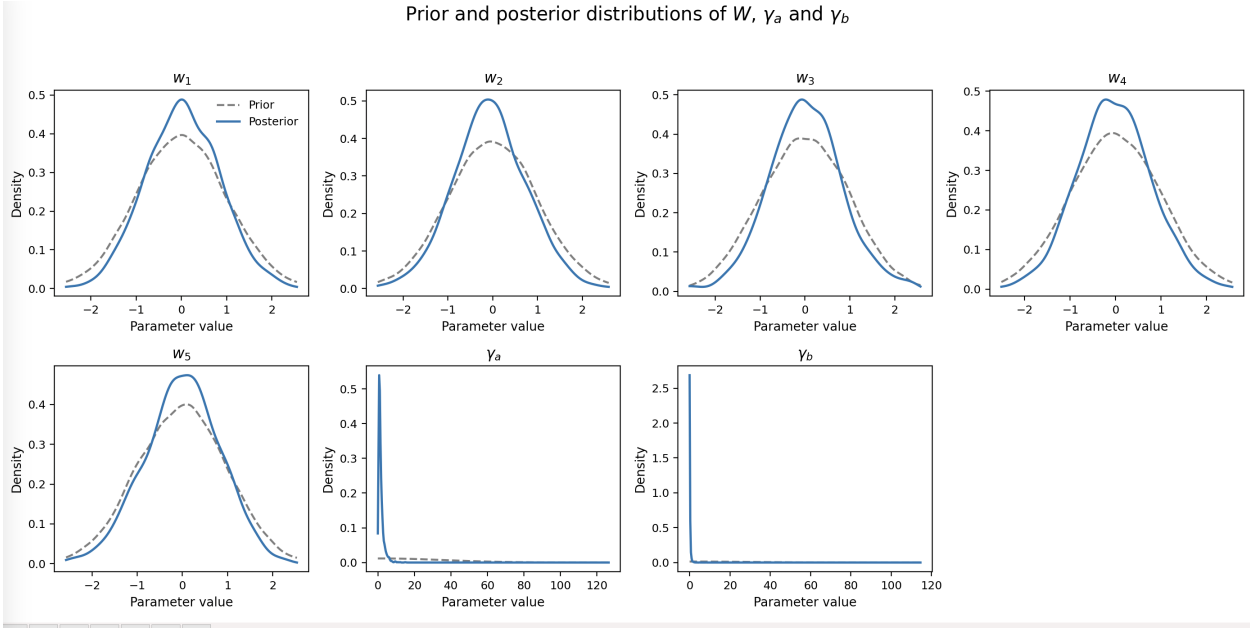


Figure 19: Prior vs Posterior

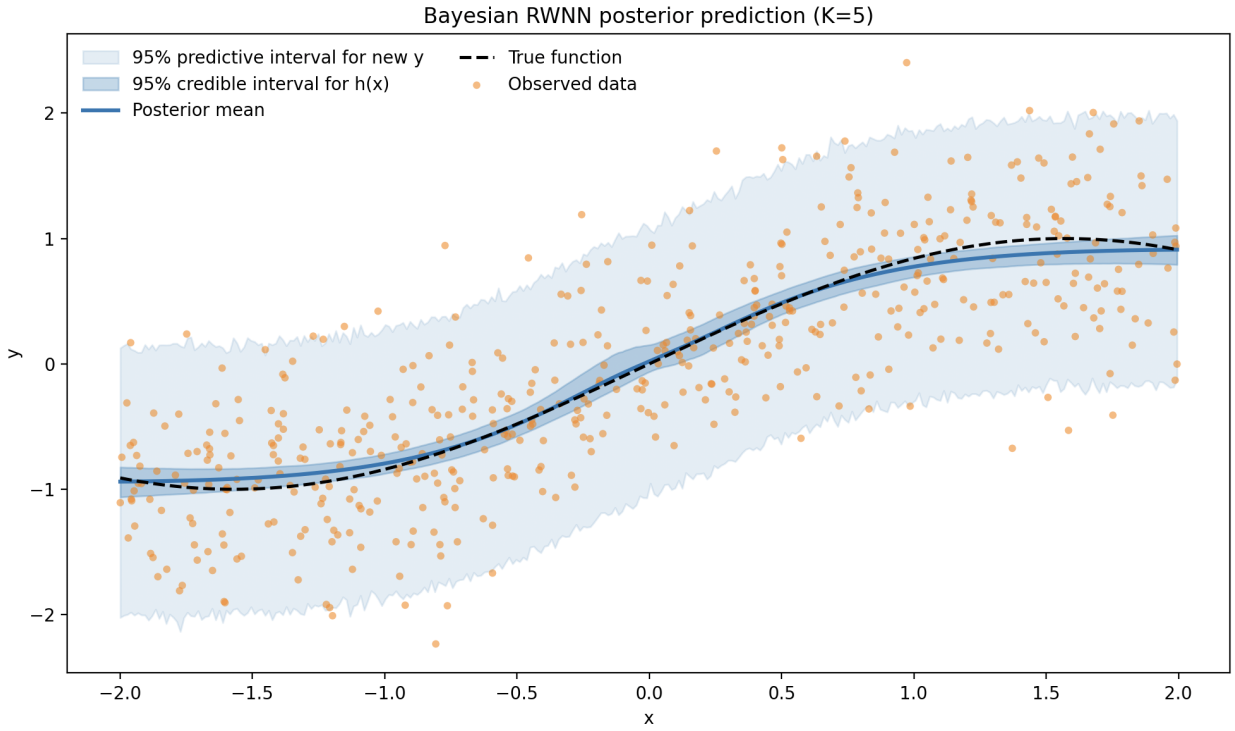


Figure 20: Prediction

We could see that γ_b is strongly centered near zero, while γ_a retains a broder posterior distribution. Since γ_a and γ_b control the variability of a_k and b_k, b_k is centered largely around 0 while a_k is more variable.

Thus $tanh(a_k x + b_k) \approx tanh(a_k x)$. This makes sense as $sin(x)$ is an odd function so we nedd an odd function to fit. This means the hierarchical model learns how to set hidden-layer parameters properly from the data.

The Stan diagnostics are satisfactory, apart from one divergent transition among 2,000 retained draws.

In terms of the prediction plot, the model provides excellent fitting between -1 and 1. For x greater than 1, the posterior mean is lower than the true function while for x less than -1, the posterior mean is greater tha the true function. When we increase K to 10, the change in prediction results is negligible.

The introduction of uncertainty in γ_a and γ_b does not appear to improve point prediction accuracy in the present experiment. Its main benefit is instead to learn the hidden-parameter scales from the data and propagate their uncertainty into posterior prediction.

6.4 Test with $h(x) = \sin(3\pi x/2)/[1 + 18x^2(\text{sgn}(x) + 1)]$

Here we set $K = 5$ to reduce running time, $\sigma_\epsilon^2 = 0.3$ and 2000 draws again.

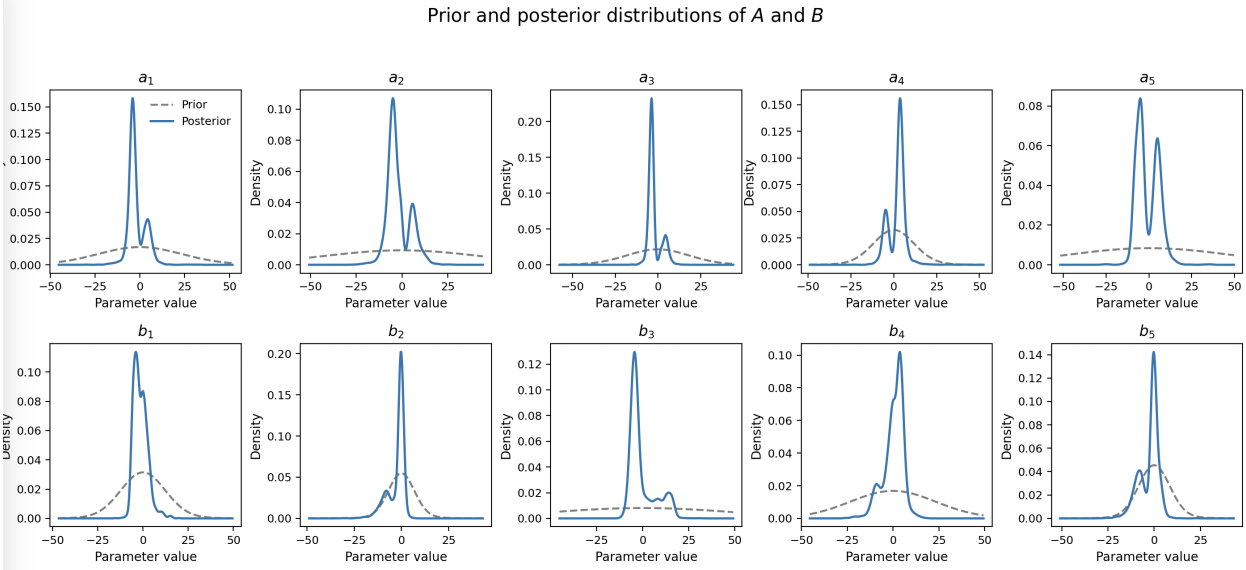


Figure 21: Prior vs Posterior

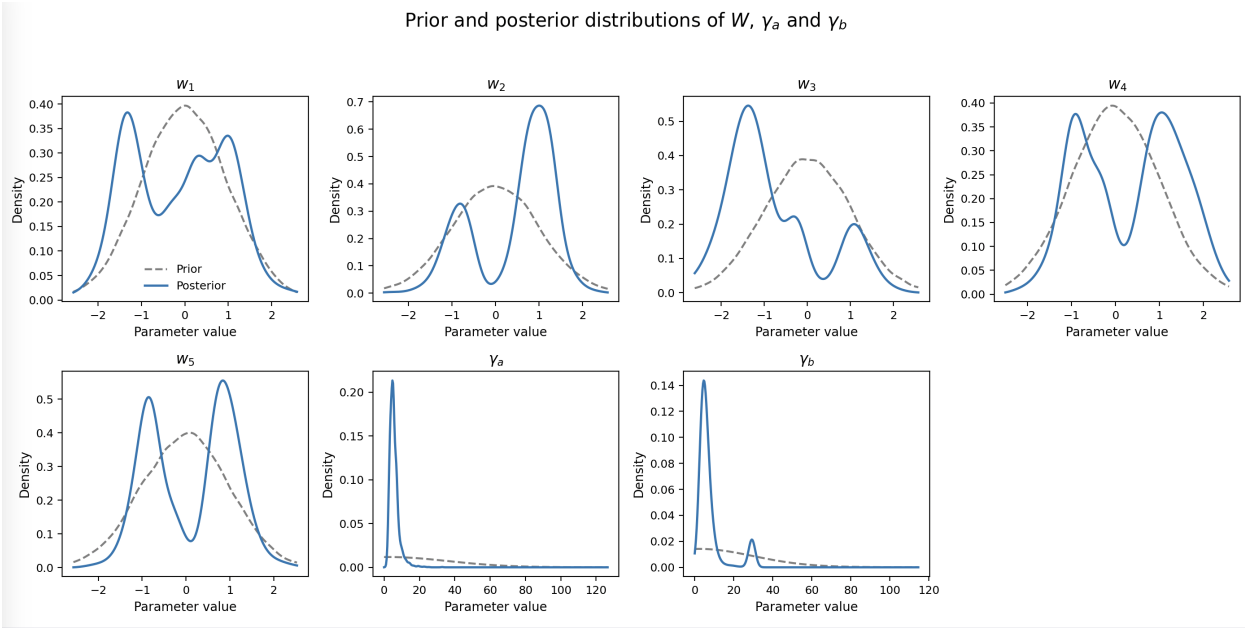


Figure 22: Prior vs Posterior

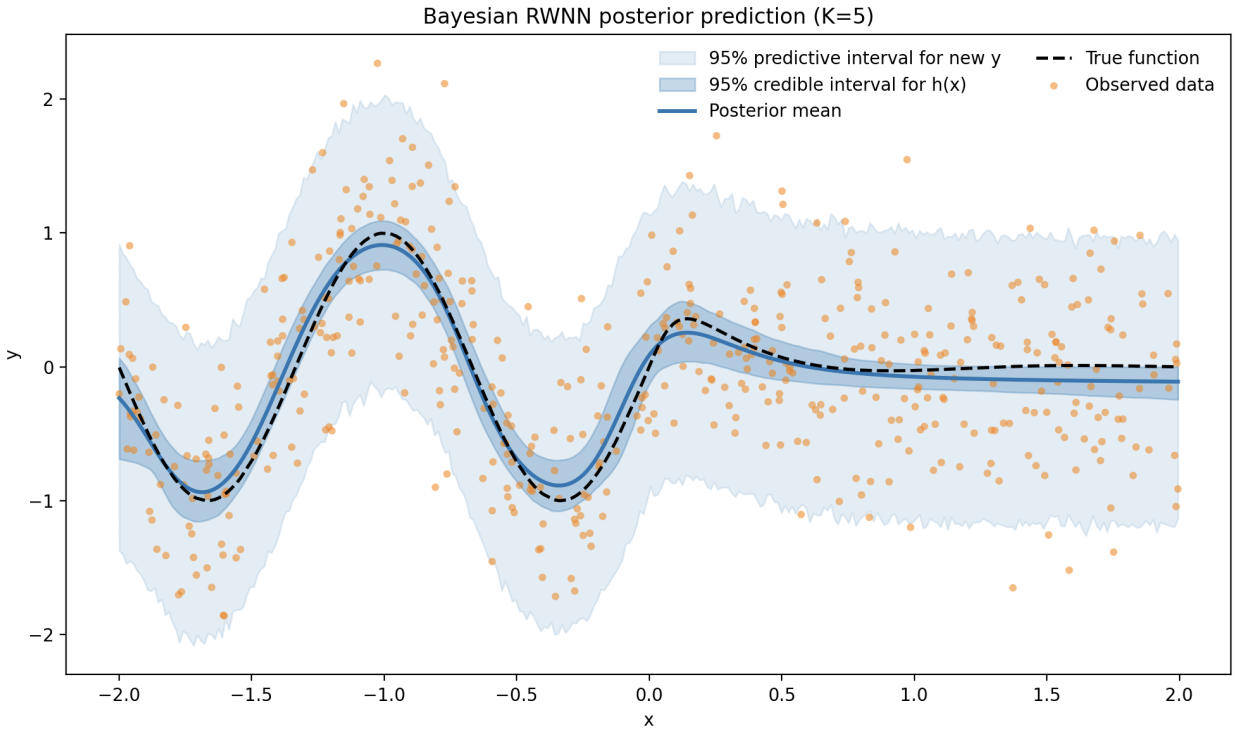


Figure 23: Prediction

6.5 Sensitivity to prior

Previously we set HalfCauchy as the hyperprior, but it may not be the best prior. Hence in this section we are going to test whether the method is sensitive to hyperprior. We would examine Half-Cauchy, Half-t(3) and Half-Normal. Here we still set $h(x) = sin(x)$ and K=5 to speed up the process.

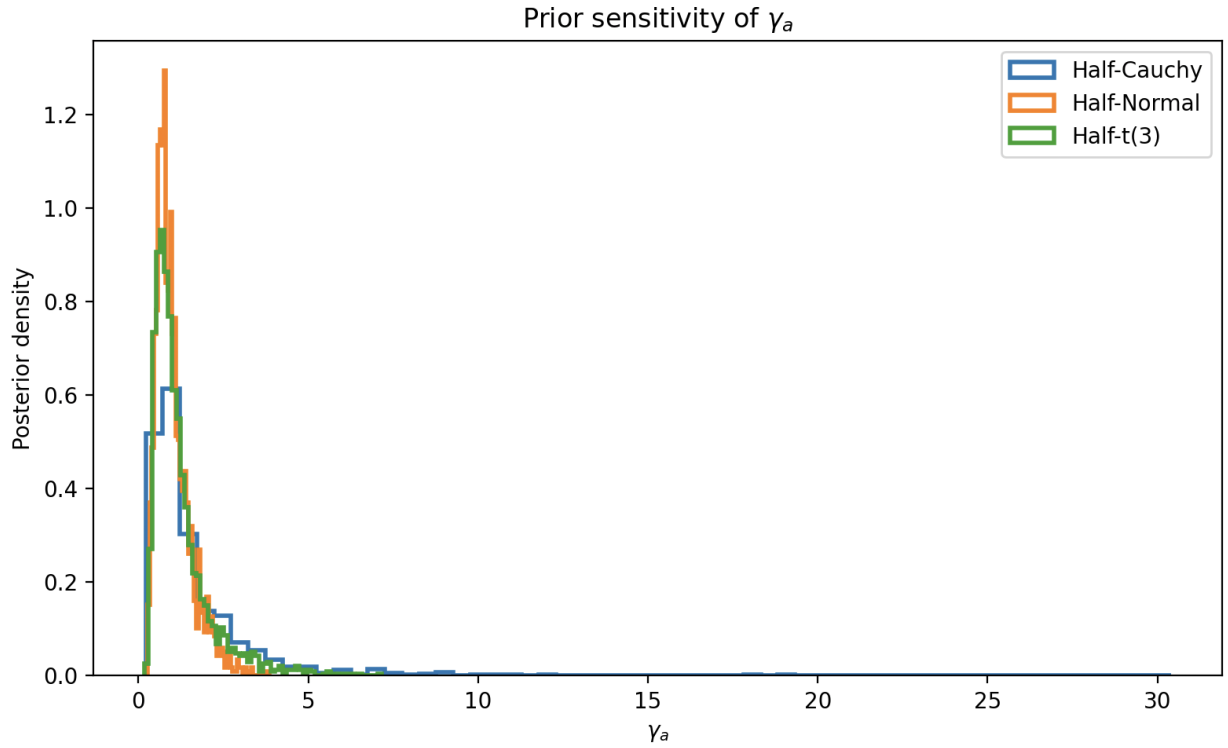


Figure 24: Comparison of γ_a

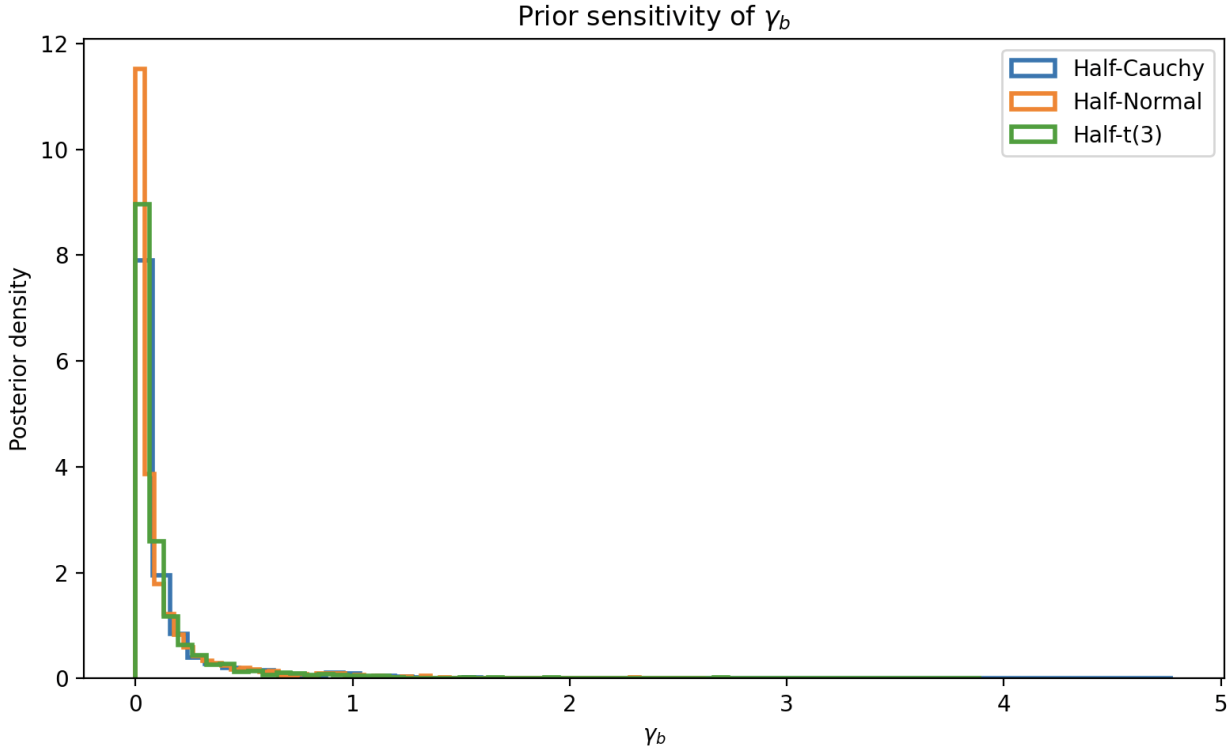


Figure 25: Comparison of γ_b

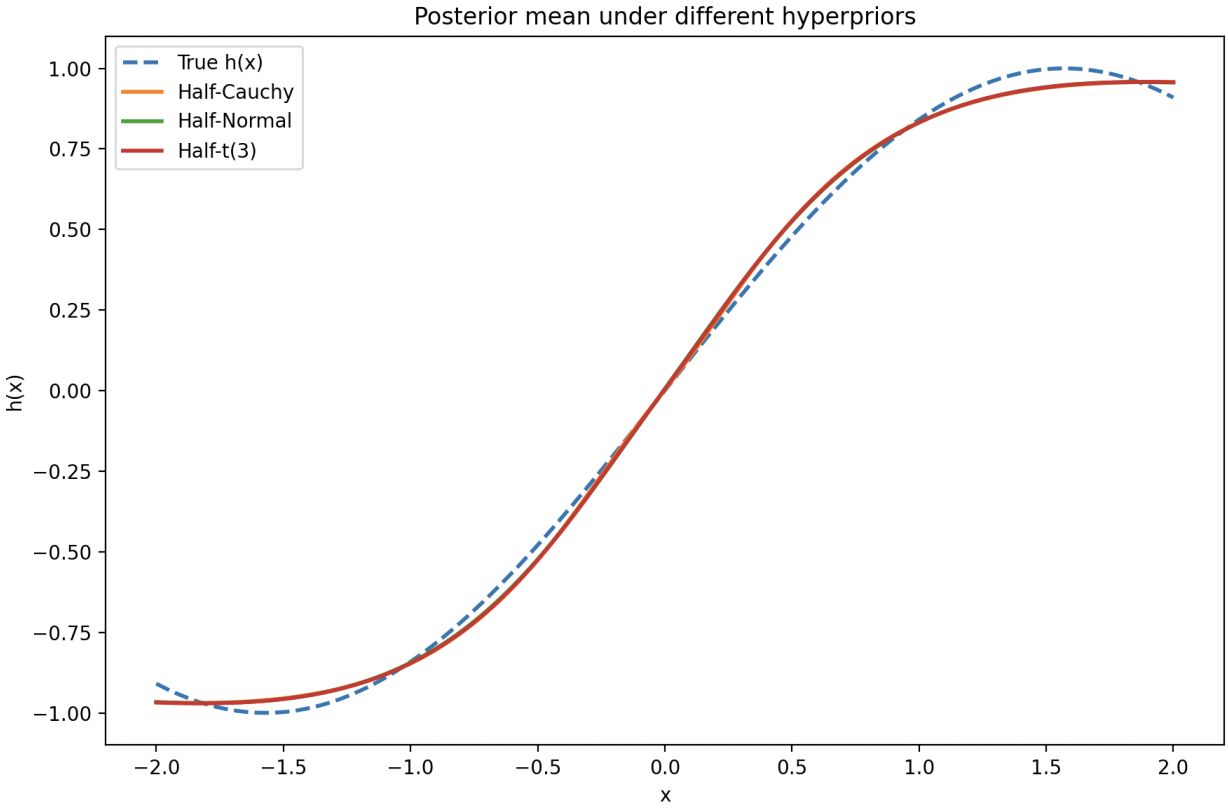


Figure 26: Comparison of h

The posterior distributions show some sensitivity to the choice of hyperprior. However, the inference for $h(x)$ is highly stable: the posterior mean functions and credible intervals(we did not put plots here) are nearly indistinguishable across hyperpriors. This suggests that the function-level inference is robust.

7 Stability of $h(x)$

7.1 Motivation

Based on the hierarchical model in the previous section, we are interested in the stability of $h(x)$. In this section, we still simulate the posterior distributions of A and B, but we will use ridge regression to obtain W, back to our original idea. In addition, we will look at the effect of K on the posterior and prediction results.

For a fixed K, suppose that S posterior samples $(A^{(s)}, B^{(s)})$, $s = 1, \ldots, S$, are obtained from the hierarchical Stan model. For each posterior draw, the output weights are evaluated by ridge regression, producing a fitted function

$$\hat{h}^{(s)}(x) = \sum_{k=1}^{K} \hat{w}_k^{(s)} \tanh\left(a_k^{(s)} x + b_k^{(s)}\right)$$

The stability of the fitted function is evaluated on a grid

$$x = 1, \ldots, x_G,$$

where $G = 300$ in the experiments.

Mean pointwise standard deviation. At each grid point x_g, the variability of the fitted function is measured by

$$\mathrm{SD}_h(x_g) = \sqrt{\frac{1}{S-1}\sum_{s=1}^{S}\left(\hat{h}^{(s)}(x_g) - \bar{h}(x_g)\right)^2}$$

, where

$$\bar{h}(x_g) = \frac{1}{S}\sum_{s=1}^{S}\hat{h}^{(s)}(x_g)$$

The overall functional variability is then summarized by averaging the pointwise standard deviations over the grid:

$$\mathrm{MeanSD}(h) = \frac{1}{G}\sum_{g=1}^{G}\mathrm{SD}_h(x_g)$$

A smaller value of $\mathrm{MeanSD}(h)$ indicates that the fitted function is less sensitive to different posterior draws of the hidden parameters.

Pairwise functional RMSE To directly compare two fitted functions obtained from different posterior draws, define the pairwise functional distance

$$D_{st} = \sqrt{\frac{1}{G}\sum_{g=1}^{G}\left[\hat{h}^{(s)}(x_g) - \hat{h}^{(t)}(x_g)\right]^2},$$

for two posterior draws s and t.

The overall pairwise functional variability is summarized as

$$\overline{D} = \frac{1}{M}\sum_{m=1}^{M} D_{s_m t_m},$$

where M randomly selected pairs of posterior draws are used. In the experiments, $M = 1000$.

7.2 Experiment with $h(x) = \sin(3\pi x/2)/[1 + 18x^2(\mathrm{sgn}(x) + 1)]$

We run four chains, each chain has 500 posterior samples, so we have 2000 samples in total.

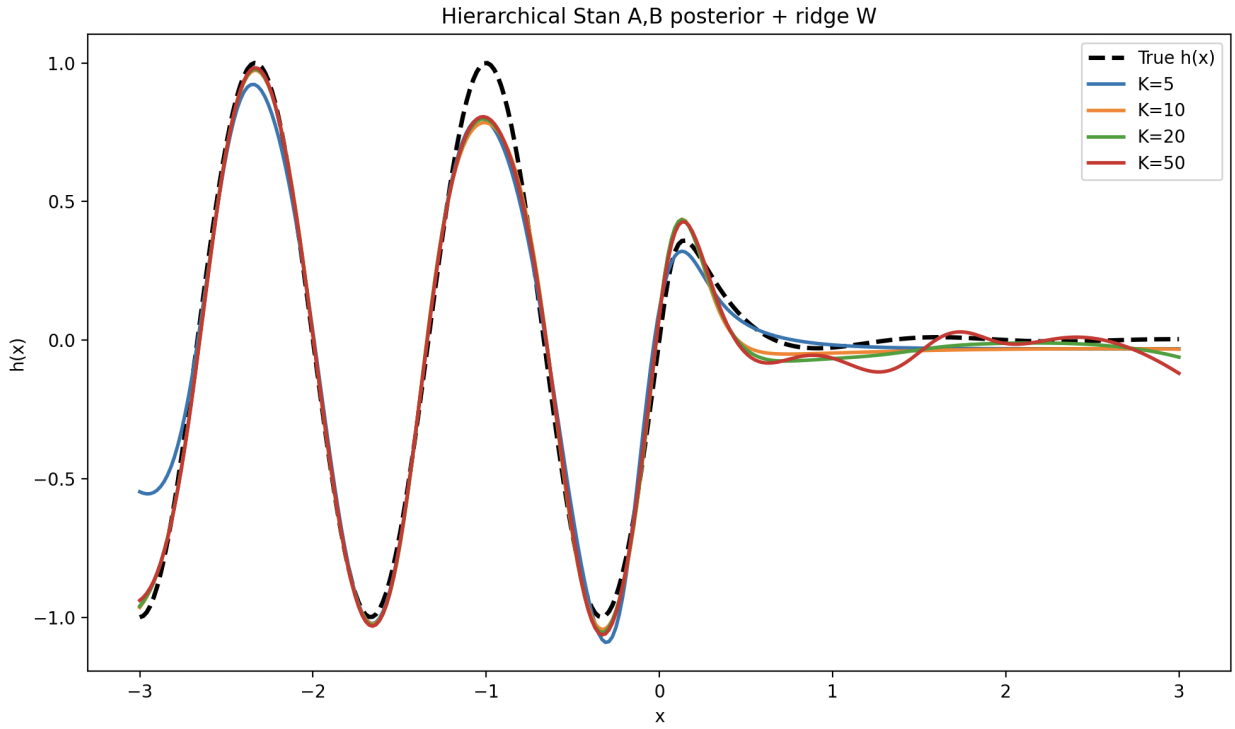


Figure 27: Comparison respect to K

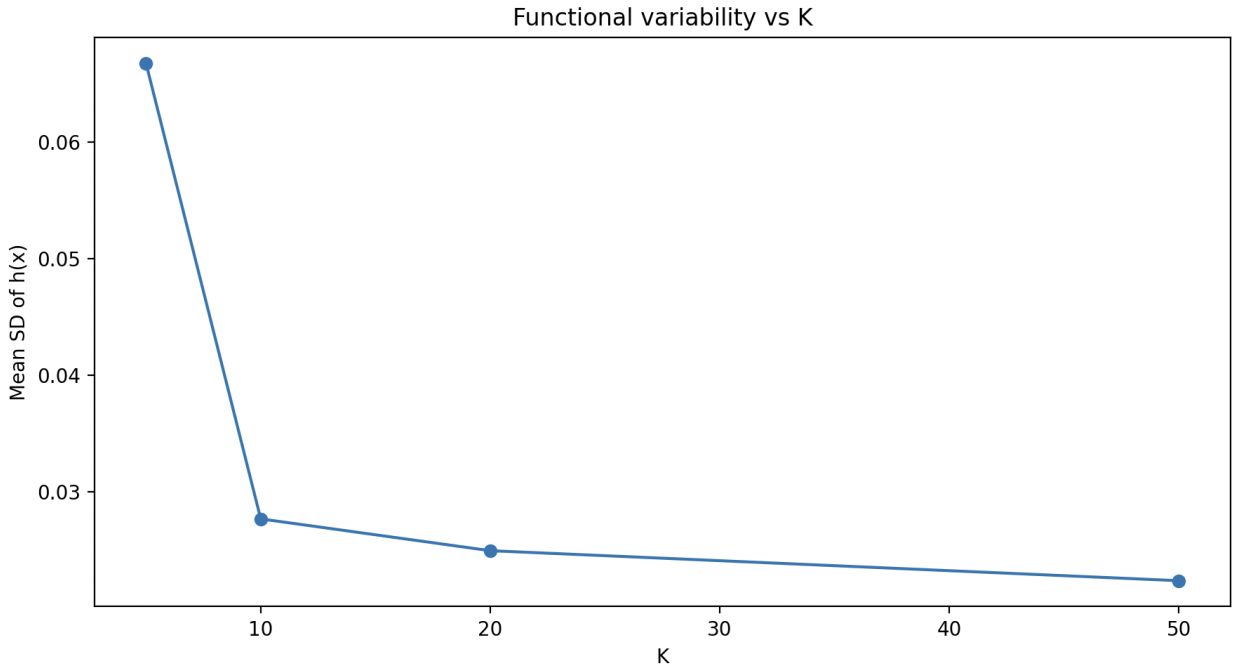


Figure 28: Mean SD

The Mean SD decreases sharply when K increases from 5 to 10, and the change is negilgible across 20 to 50.

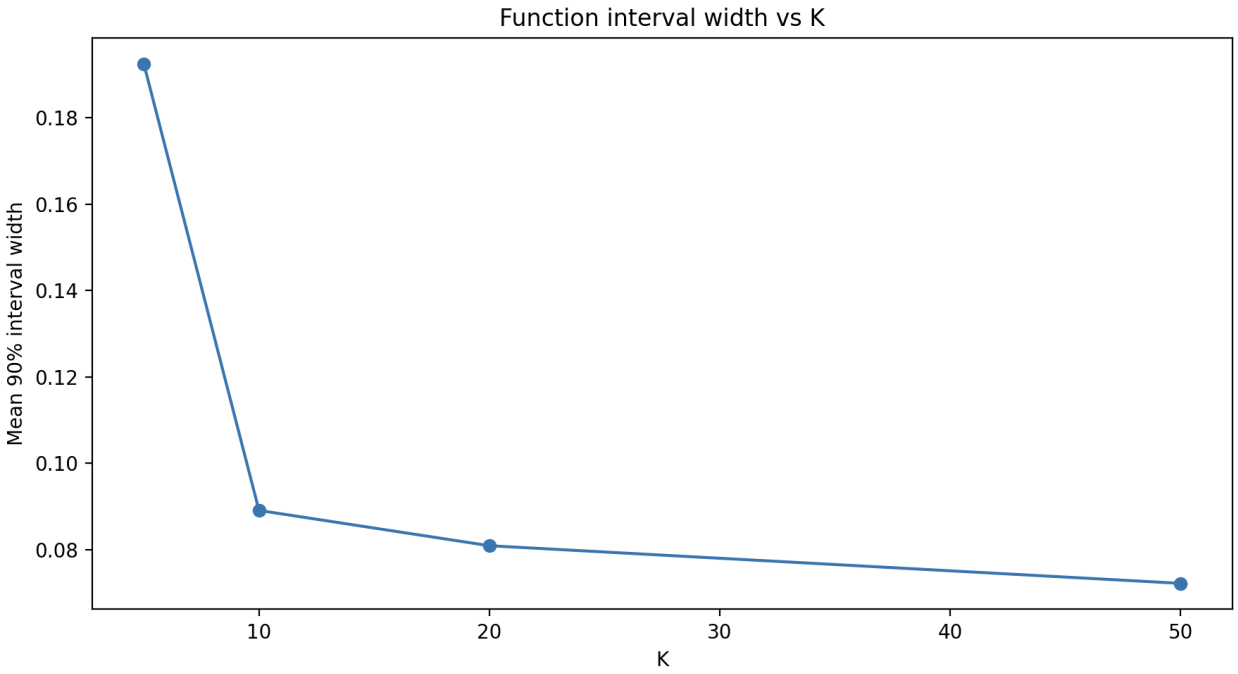


Figure 29: Interval width

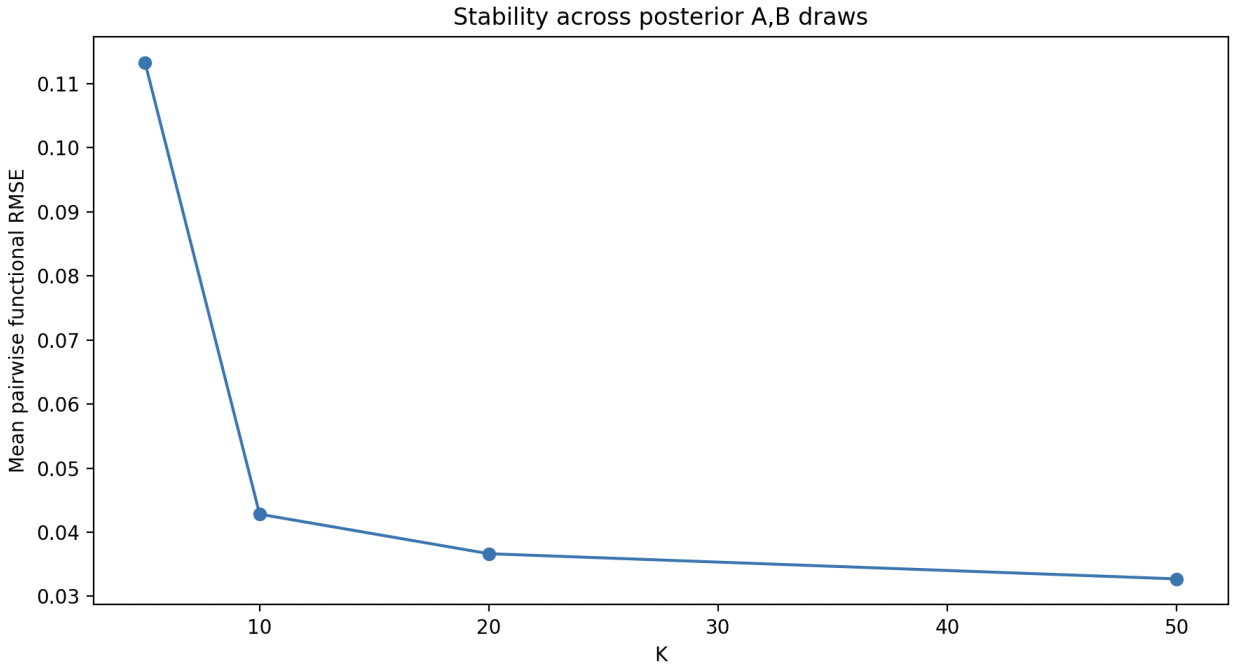


Figure 30: Pairwise RMSE

The same trend is witnessed in the other two plots.
Finally, we could look at the run time.

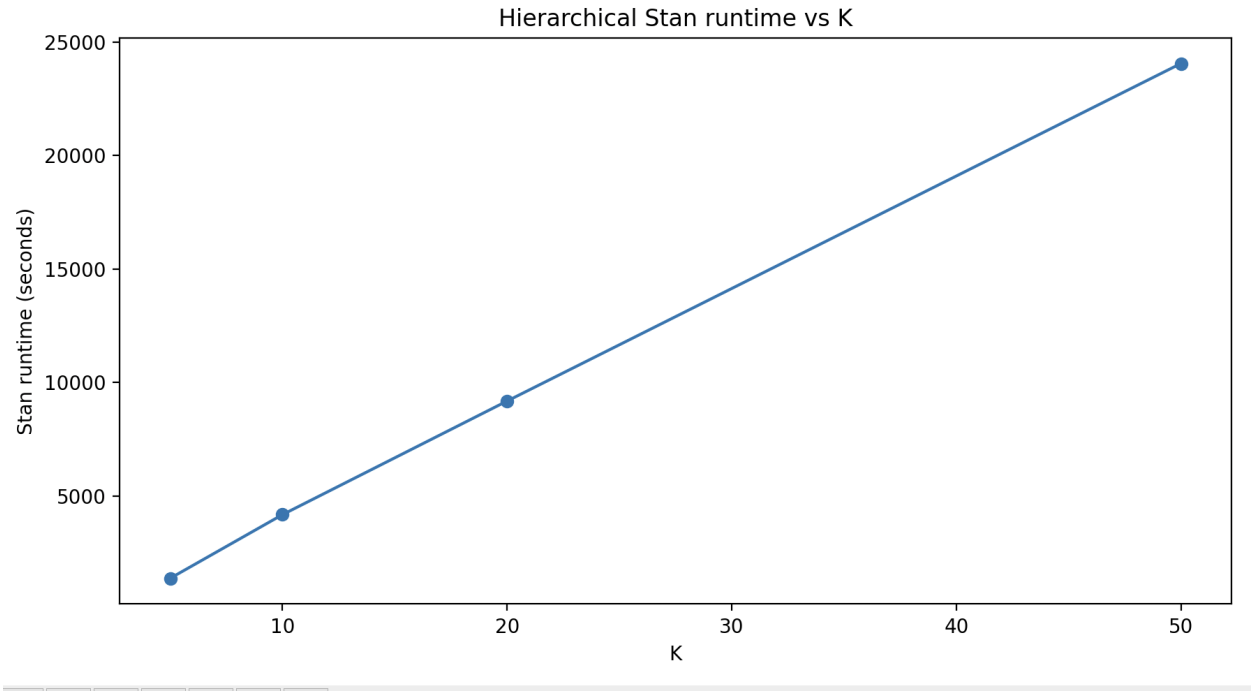


Figure 31: Running time

It takes more than one hour to run the result when K=10, for K=50, the running time is so long!

Also, the parameter convergence is poor for K=20 and K=50. Hence K=10 might be a wise choice.

Then we think of reducing sampling to make this process quicker. Now we still set four chains but we set 250 samples for each chain. And we drop the K=50 case.

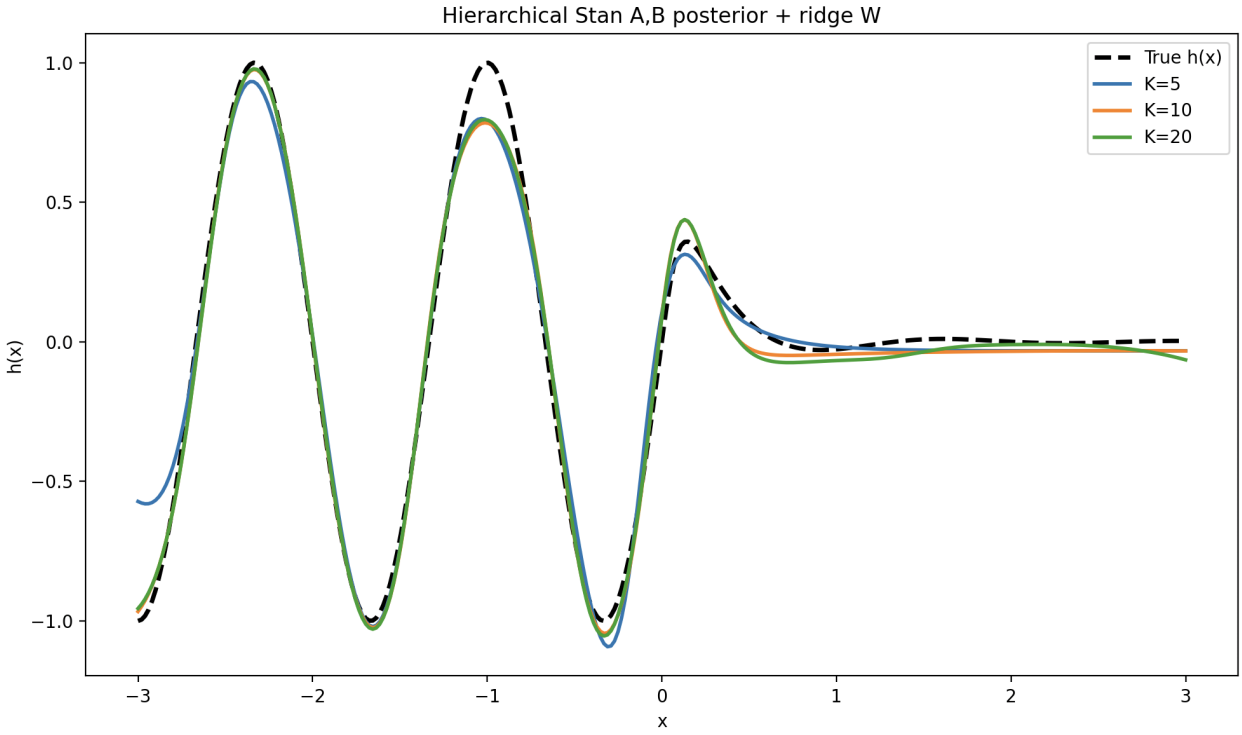


Figure 32

This plot shows that fewer posterior samples could also give good fit for $h(x)$. The test RMSE is similar to that of 2000 samples(both lower than 0.10 for each K).

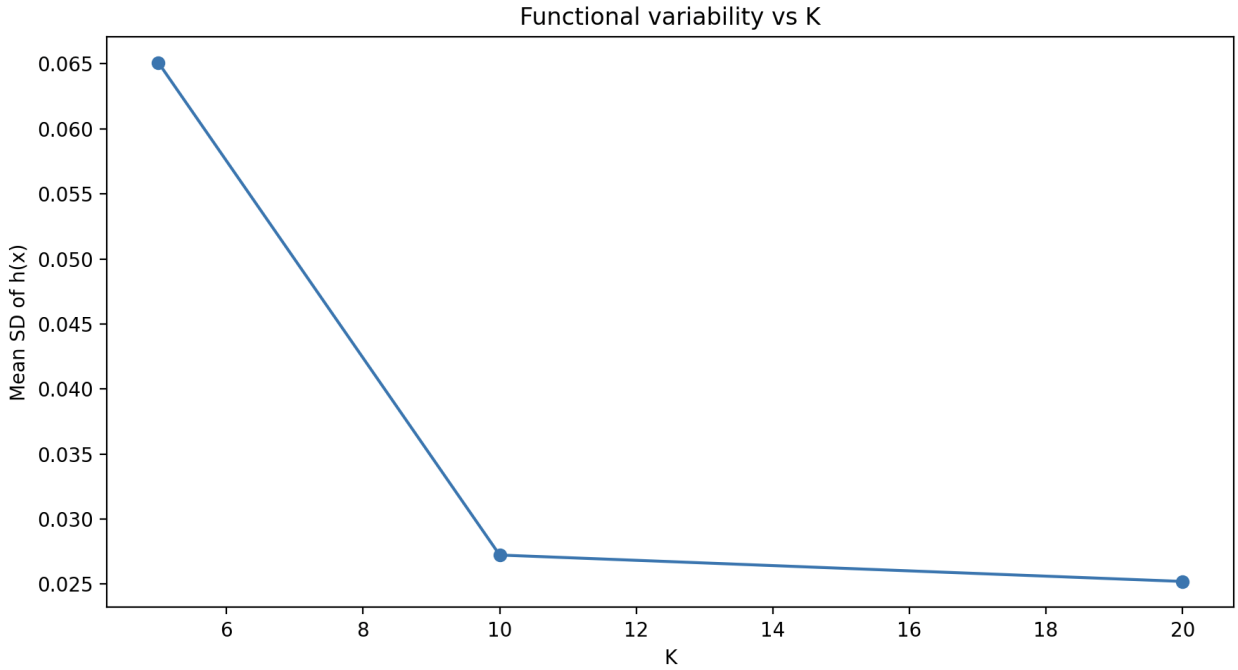


Figure 33: Mean SD 1000 samples

Again, a large decrease is witnessed when K increases from 5 to 10.

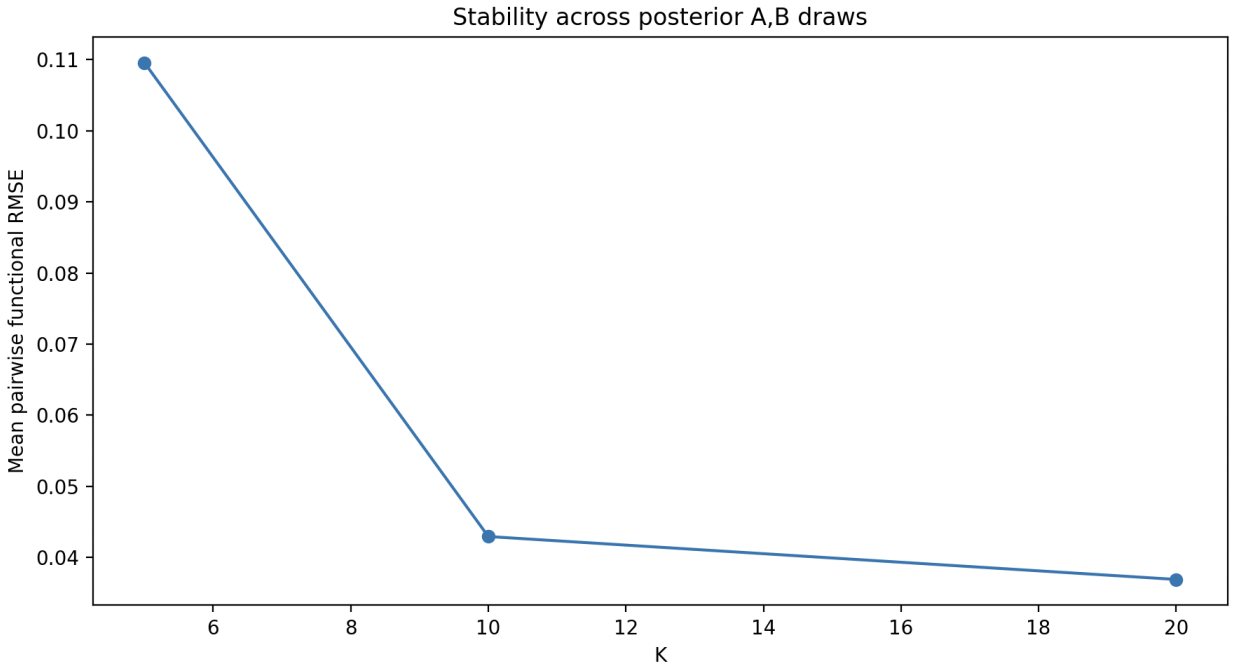


Figure 34: Pairwise functional RMSE 1000 samples

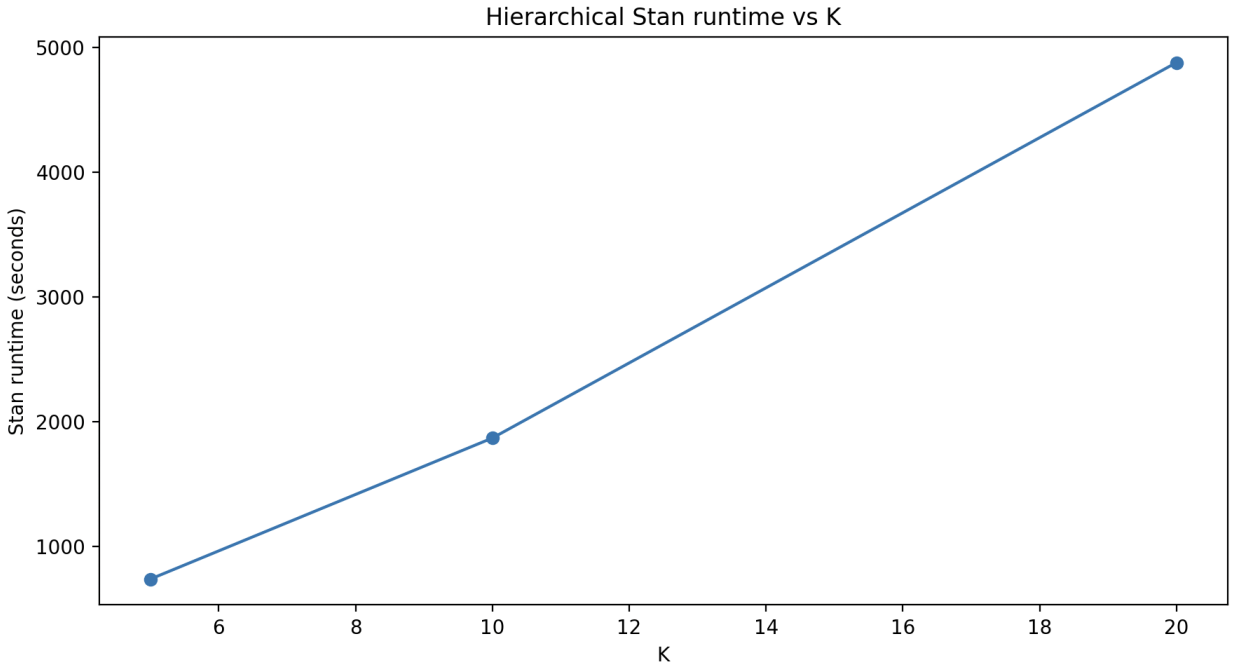


Figure 35: Running time 1000 samples

Based on the results above, we found that when K increases from 10 to 20, the improve-

ment in fitting for $h(x)$ is samll but the running time explodes. Thus K = 10 should be a relatively safer choice.

However, Stan suggests that the parameter convergence is poor for K=10, as well. 777 of 1000 transitions hit the maximum treeepth limit. This could be partly explained by the non-identifiability of our model.

We could then investigate whether the four chains give the similar fitted function. If the answer is yes, then the stability of $h(x)$ could be shown.

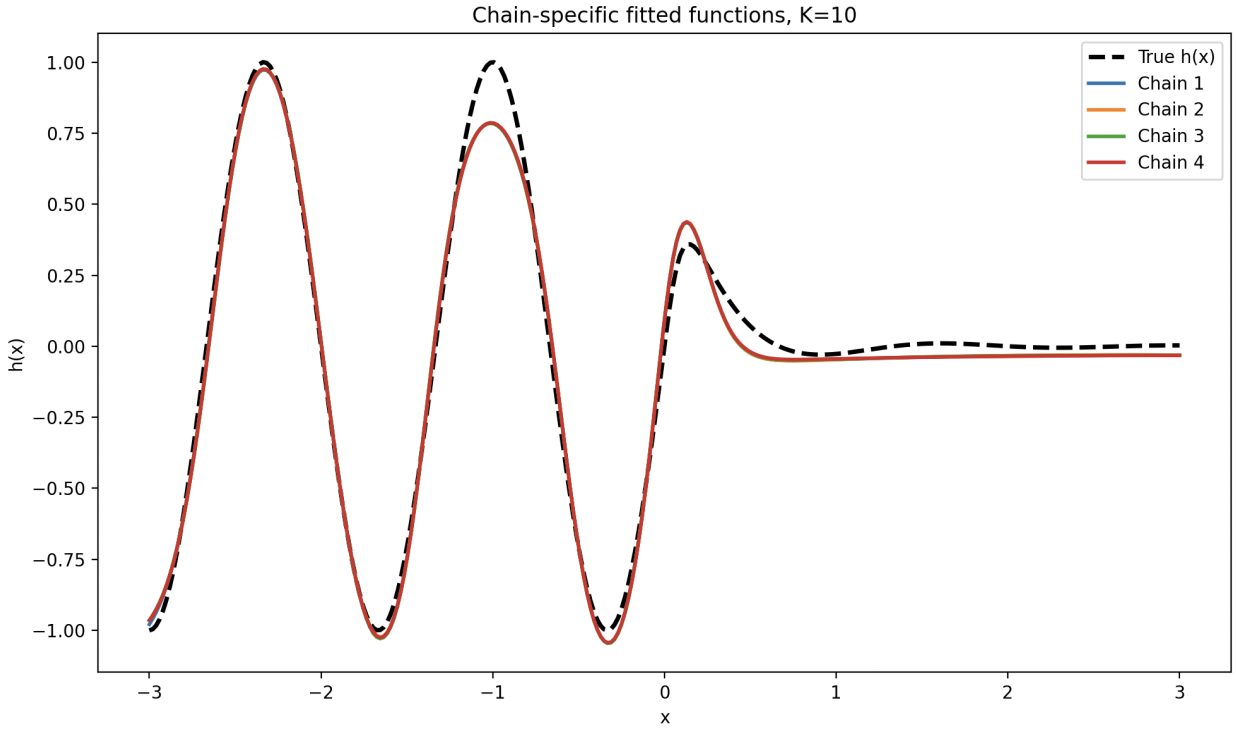


Figure 36: Comparison across chains, K=10

The four fitted functions almost overlap! This could lead to an interesting observation: the non-identifiability of parameters does not translate into the instability of fitted function.

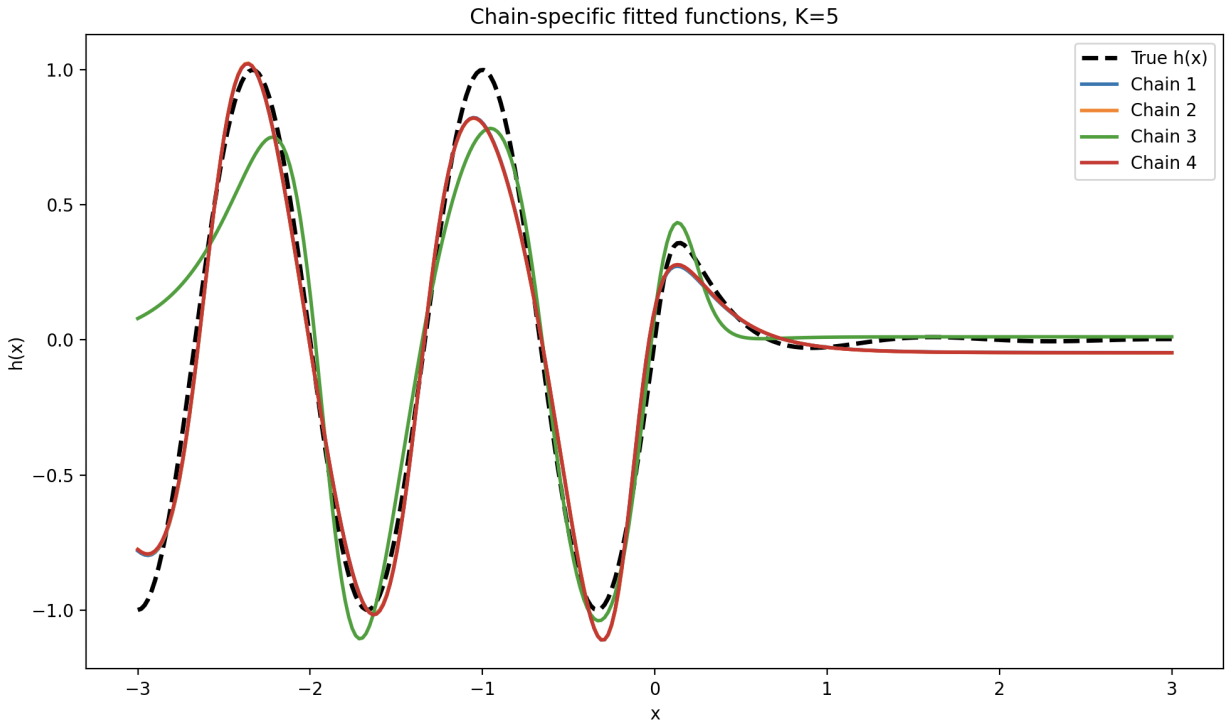


Figure 37: Comparison across chains, K=5

When K=5, three chains produce almost overlapping fitted functions, while one chain exhibits noticeable deviations. But the functional variability across chains is much smaller than parameter-level variability.

7.3 Further Check of Stability

Previously we only look at the mean of chain-specific fitted functions. In this section we will check stability of the fitted functions more carefully.

For each retained iteration of each Stan chain, we construct functions using the same hidden parameters (A, B). The first uses ridge readout weights while the second uses the corresponding joint Stan draw of W, thereby retaining its uncertainty. This paired comparison allows us to examine how replacing sampled readout weights by their conditional posterior mean affects both the fitted function and its variability.

We evaluate $\widehat{R}$, bulk and tail effective sample sizes, and Monte Carlo standrd errors(MCSE) for the function values on an evaluation grid. We also assess three function-level quantities: the global peak, the signed integral, and the test RMSE relative to the true function.

In the following experiment, we fix K=10. The same function is used as the previous section.

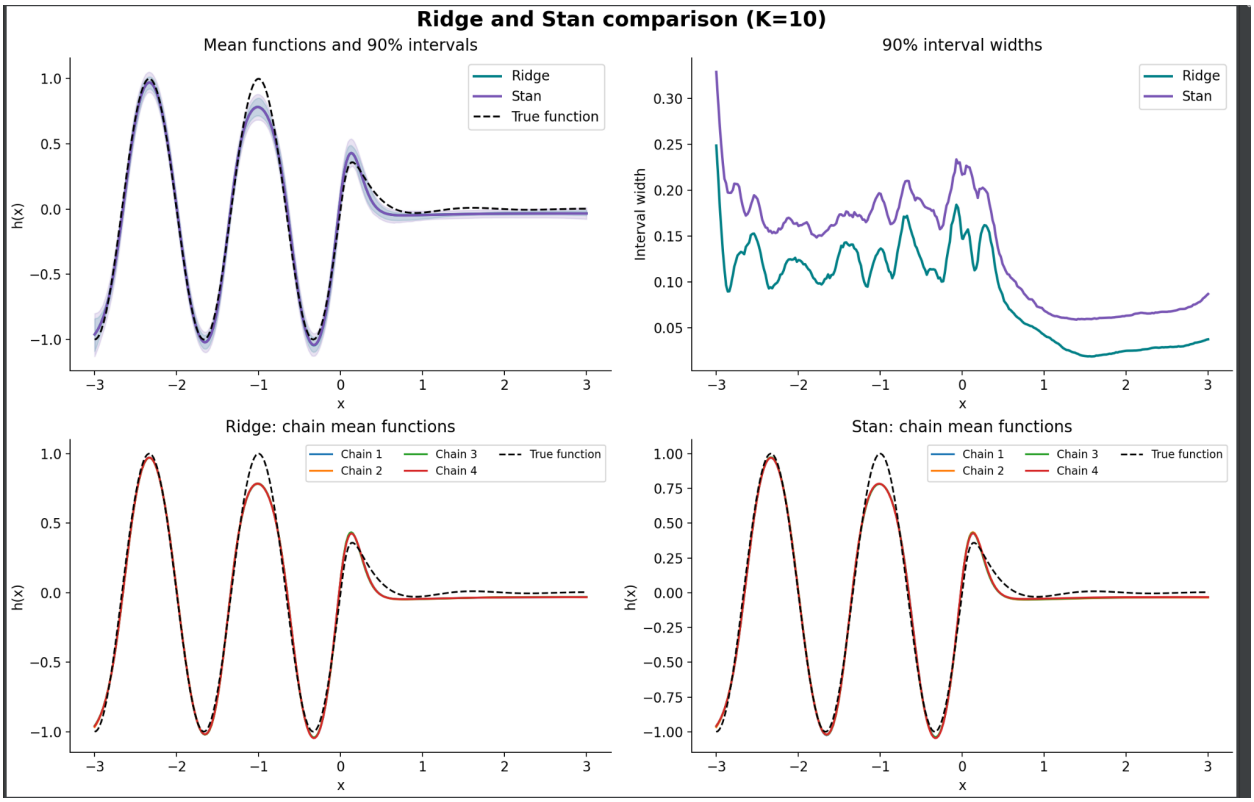


Figure 38: Comparison across ridge and Stan

The mean fitted functions of Ridge and Stan almost overlap. However, the interval widths of Stan is higher. This is as expected since Stan preserves the uncertainty of W.

Function stability summary (K=10)

Function variability and prediction accuracy

Method	Mean SD	Mean 90% width	Grid RMSE vs truth	Test RMSE vs truth	Test RMSE vs noisy Y
Ridge	0.0275	0.0884	0.0579	0.0534	0.2866
Stan	0.0415	0.1358	0.0580	0.0536	0.2867

Worst pointwise diagnostics across the evaluation grid

Method	Max R-hat	Worst x (R-hat)	Min bulk ESS	Min tail ESS	Max MCSE (mean)	Max MCSE/SD	R-hat >=1.01 (points)	Nonfinit (points
Ridge	1.0159	-3.000	488	485	0.00242	0.0466	9	0
Stan	1.0070	-2.699	690	500	0.00310	0.0388	0	0

Diagnostics for function-level quantities

Method	Quantity	Mean	SD	R-hat	Bulk ESS	Tail ESS	MCSE (mean)	MCSE/S
Ridge	Global peak	0.9734	0.0281	1.0002	936	852	0.00092	0.0329
Ridge	Integral (signed area)	-0.2684	0.0040	1.0022	986	955	0.00013	0.0322
Ridge	Test RMSE vs truth	0.0613	0.0072	1.0006	915	874	0.00024	0.0337
Stan	Global peak	0.9741	0.0465	1.0022	962	897	0.00151	0.0324
Stan	Integral (signed area)	-0.2699	0.0654	1.0002	1010	884	0.00208	0.0318
Stan	Test RMSE vs truth	0.0690	0.0099	0.9999	831	944	0.00035	0.0351

Stan sampler diagnostics

Chain	Retained draws	Divergences	Max-depth hits	E-BFMI
1	250	0	213	0.974
2	250	0	231	0.899
3	250	0	187	0.934
4	250	0	186	0.875

Figure 39: Summary Statistics

Diagnostic Criteria For the four-chain experiments, we adopt $\widehat{R} < 1.01$ and bulk and tail ESS of at least 400 as diagnostic targets(suggested by Stan). For posterior means, we additionally require the ratio of MCSE to posterior standard deviation to be below 0.05.

These criteria are applied both to function values on the evaluation grid and to the selected function-level quantities. We report the worst pointwise diagnostics to identify local sampling difficulties that may be concealed by averaging. At the sampler level, we check for divergent transitions, low E-BFMI (below 0.3), and frequent saturation of the maximum tree depth.

Analysis The function-level sampling diagnostics are generally satisfactory. For the joint Stan functions, all grid points have $\widehat{R} < 1.01$, with a maximum of 1.0070. The minimum bulk and tail ESS are 690 and 500 respectively. For ridge, the corresponding minima are 488 and 485, but 9 grid points have $\widehat{R} \geq 1.01$.

The global peak, signed integral, and test RMSE vs truth also show satisfactory sampling consistency under both constructions. Their $\widehat{R}$ values are less than 1.01, bulk and tail ESS all exceed 800. These results support the stability of function-level quantities.

However, sampling efficiency is the major concern: 817 of the 1000 retained transitions

reach the maximum tree death.

8 Hierarchical TAGI

8.1 Motivation

Based on the previous section regarding the function-level stability, we could see the consistency of our sampling method. However, Stan sampling is inefficient. The main objective of our project is to sample quickly from the posterior. TAGI is fast but it is operated under fixed scale(γ_a and γ_b). Therefore it is natural to come up with a new algorithm that combines TAGI and the hierarchical model together!